INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Topic 08: Master ITM Papers 01 to 07 Full Exam Simulations

Section: 09_Mock_Test_Series | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Topic 08: Master ITM Papers 01 to 07 Full Exam Simulations

Complete 60-Question Full-Length Mock Exams for Timed Simulation

> **Exam Blueprint:** 60 Questions | 120 Minutes Total | Sectional Time Limits Apply
> **Sections:**
> - Section 1: Reasoning Ability (Q1–Q15, 25 min)
> - Section 2: Technical Ability, Quant (Q16–Q25, 35 min)
> - Section 3: Verbal Ability (Q26–Q45, 20 min)
> - Section 4: Pseudo Code (Q46–Q50, 10 min)
> - Section 5: Numeric Puzzle Solving (Q51–Q54, 10 min)
> - Section 6: English Grammar (Q55–Q59, 10 min)
> - Section 7: English Writing (Q60, 10 min)

Mock Exam Paper 01: Infosys System Engineer ITM Mock Paper 1

Source File: `Infosys_SE_Mock_Paper.pdf`
Total Questions: 60 Questions | **Total Marks:** 69 Marks (59 MCQs + 10 Marks Writing)

SECTION 1: Reasoning Ability (25 min)

Q01. Statements: All engineers are coders. Some coders are testers. No tester is a manager.
Conclusions: I. Some coders are not managers. II. Some engineers are testers. III. No engineer is a manager.
(a) Only I follows
(b) I and II follow
(c) I and III follow
(d) All follow
Answer: **(A)** | *Explanation:* Statement I: All engineers are coders, Some coders are testers, No tester is a manager. The subset of coders who are testers cannot be manag...

Q02. Statements: Some pens are books. All books are chairs. No chair is a table.
Conclusions: I. Some pens are chairs. II. No book is a table. III. Some pens are not tables. IV. Some tables are pens.
(a) I and II only
(b) I, II and III only

(c) II, III and IV only
(d) All follow
Answer: **(B)** | *Explanation:* Statements: Some pens are books. All books are chairs. No chair is a table.
- Conclusion I: Some pens are chairs. Since some pens are books ...

Q03. Data Sufficiency. For integers x and y, is x > y?
I. $x^2 > y^2$. II. x + y > 0.
(a) I alone is sufficient
(b) II alone is sufficient
(c) Both together are sufficient, neither alone
(d) Even both
together are not sufficient
Answer: **(C)** | *Explanation:* Question: Is x > y for integers x and y?
- Statement I: $x^2 > y^2$ gives |x| > |y|. If x = -3, y = 2, $x^2 = 9 > 4$ but x < y. If x = 3, y = 2, x ...

Q04. Data Sufficiency. What is Riya's present age?
I. Riya is twice as old as her sister, who is 3 years younger than their brother.
II. Five years hence, the sum of the ages of Riya and her sister will be 40.
(a) I alone is sufficient
(b) II alone is sufficient
(c) Both together are sufficient, neither alone
(d) Even both
together are not sufficient
Answer: **(C)** | *Explanation:* Question: What is Riya's present age?
- Statement I: R = 2S, S = B - 3. Three unknowns, two equations. Insufficient.
- Statement II: (R + 5)...

Q05. In a certain code, MONDAY is written as NNOCBX. How is SUNDAY written in that code?
(a) TTOCBX
(b) TVOEBZ
(c) TTOCBZ
(d) RTOCBX
Answer: **(A)** | *Explanation:* Coding Pattern: M (+1) -> N, O (-1) -> N, N (+1) -> O, D (-1) -> C, A (+1) -> B, Y (-1) -> X. The alternating shift pattern is [+1, -1, +1, ...

Q06. In a code, "sun rises early" = "ka mo ti", "early bird catches" = "ti ra po", and "bird sun light" = "ra ka du". What is the code for "rises bird light"?
(a) mo ra du
(b) mo po du
(c) ka ra du
(d) mo ti du
Answer: **(A)** | *Explanation:* Word-code comparison:
1) 'sun rises early' = 'ka mo ti'
2) 'early bird catches' = 'ti ra po' => Common word 'early' = 'ti'.
3) 'bird sun lig...

Q07. A solid cube painted red on all faces is cut into 64 identical small cubes (4 × 4 × 4). How many small cubes have exactly two faces painted?

(a) 12
(b) 24
(c) 36
(d) 8
Answer: **(B)** | *Explanation:* For an n × n × n painted cube cut into unit cubes (here n = 4 since $4^3 = 64$):
Cubes with exactly two faces painted lie along the 12 edges (e...

Q08. In triangle ABC, the medians AD, BE and CF are drawn and meet at G. How many triangles are there in the figure in total?
(a) 12
(b) 14
(c) 16
(d) 18
Answer: **(C)** | *Explanation:* Standard median geometry counting:
In triangle ABC with medians AD, BE, CF intersecting at centroid G:
1) 6 small elementary triangles forme...

Q09. Pointing to a man, Meera said, "His mother is the only daughter of my mother." How is Meera related to the man?
(a) Mother
(b) Sister
(c) Aunt
(d) Grandmother
Answer: **(A)** | *Explanation:* Meera said: 'His mother is the only daughter of my mother.'
The 'only daughter of my mother' when spoken by a woman (Meera) refers to Meera ...

Q10. If A + B means A is the mother of B, A − B means A is the brother of B, and A × B means A is the father of B, which expression shows that P is the maternal uncle of R?
(a) P + Q − R
(b) P − Q + R
(c) P × Q − R
(d) P − Q × R
Answer: **(B)** | *Explanation:* P must be the maternal uncle of R (brother of R's mother).
- Q + R means Q is the mother of R.
- P - Q means P is the brother of Q.
Combinin...

Q11. Ravi walks 10 m north, turns right and walks 6 m, turns right and walks 14 m, turns left and walks 3 m, then turns left and walks 4 m. Where is he now relative to the starting point?
(a) 9 m East
(b) 9 m West
(c) 11 m South-East
(d) 5 m North
Answer: **(A)** | *Explanation:* Step-by-step vector tracking from origin (0, 0):
1) Walks 10 m North -> (0, 10)
2) Turns right (East) and walks 6 m -> (6, 10)
3) Turns righ...

Q12. A person facing North-East turns 90° clockwise, then 135° anticlockwise, then 180° clockwise. Which direction is he facing now?
(a) North
(b) South
(c) South-West
(d) East
Answer: **(B)** | *Explanation:* Initial direction: North-East (45° clockwise from North).
1) +90° clockwise: 45° + 90° = 135° (South-East).
2) -135° anticlockwise: 135° - 1...

Q13. Find the next term: 3, 7, 16, 35, 74, ?
(a) 143
(b) 153
(c) 149
(d) 158
Answer: **(B)** | *Explanation:* Series: 3, 7, 16, 35, 74, ?
Pattern: ×2 + k where k increments by 1 at each step:
- 3 × 2 + 1 = 7
- 7 × 2 + 2 = 16
- 16 × 2 + 3 = 35
- 35 × ...

Q14. Find the next term: 4, 9, 25, 49, 121, 169, ?
(a) 289
(b) 259
(c) 361
(d) 323
Answer: **(A)** | *Explanation:* Series: 4, 9, 25, 49, 121, 169, ?
Notice the base numbers:
- $2^2 = 4$
- $3^2 = 9$
- $5^2 = 25$
- $7^2 = 49$
- $11^2 = 121$
- $13^2 = 169$
These are the squar...

Q15. Find the next letter: B, E, I, N, T, ?
(a) A
(b) Z
(c) B
(d) Y
Answer: **(A)** | *Explanation:* Letter positions in alphabet: B(2), E(5), I(9), N(14), T(20), ?
Differences:
- 5 - 2 = +3
- 9 - 5 = +4
- 14 - 9 = +5

- 20 - 14 = +6
Next dif...

SECTION 2: Technical Ability (Quantitative) (35 min)

Q16. In how many ways can the letters of COMMITTEE be arranged so that no two vowels are adjacent?
(a) 5400
(b) 3600
(c) 10800
(d) 2700
Answer: **(A)** | *Explanation:* Word COMMITTEE contains 9 letters: C(1), O(1), M(2), I(1), T(2), E(2).
Vowels: O, I, E, E (4 vowels, with two E's).
Consonants: C, M, M, T, ...

Q17. A committee of 5 is to be formed from 5 boys and 4 girls with at least 2 boys and at least 2 girls. In how many ways can this be done?
(a) 90
(b) 100
(c) 105
(d) 120
Answer: **(B)** | *Explanation:* Committee of 5 from 5 boys and 4 girls with at least 2 boys and at least 2 girls.
Possible compositions (Boys, Girls):
- Case 1: 2 Boys and ...

Q18. Two fair dice are thrown. Given that at least one die shows 4, what is the probability that the sum is a prime number?
(a) 4/11
(b) 1/3
(c) 5/11
(d) 4/9
Answer: **(A)** | *Explanation:* Two fair dice thrown. Total outcomes with at least one 4:
Sample space S = {(4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (1,4), (2,4), (3,4), (...

Q19. A bag has 5 red, 4 blue and 3 green balls. Three balls are drawn at random without replacement. What is the probability that all three are of different colours?
(a) 3/11
(b) 1/4
(c) 2/11
(d) 5/22
Answer: **(A)** | *Explanation:* Total balls = 5 red + 4 blue + 3 green = 12 balls.
3 balls drawn at random without replacement.
Total ways = C(12, 3) = (12 × 11 × 10) / (3 ...

Q20. A shopkeeper marks his goods 40% above cost price and allows two successive discounts of 10% and 15%. His net profit percentage is:
(a) 7.1%
(b) 5.5%
(c) 8.6%

(d) 6.4%
Answer: **(A)** | *Explanation:* Let Cost Price (CP) = 100.
Marked Price (MP) = 100 × 1.40 = 140.
Successive discounts of 10% and 15%:
Selling Price (SP) = 140 × (1 - 0.10) ...

Q21. A, B and C can individually finish a job in 12, 18 and 24 days respectively. They work on alternate days in the order A, B, C, A, B, C, … (one person per day, starting with A). On which day is the job completed?
(a) 16th
(b) 17th
(c) 18th
(d) 19th
Answer: **(B)** | *Explanation:* Let total work = LCM(12, 18, 24) = 72 units.
Daily efficiencies:
- A = 72 / 12 = 6 units/day
- B = 72 / 18 = 4 units/day
- C = 72 / 24 = 3 u...

Q22. Two trains of lengths 240 m and 160 m run on parallel tracks. Running in opposite directions they cross each other in 8 s; running in the same direction the faster overtakes the slower in 40 s. The speed of the faster train is:
(a) 90 km/h
(b) 108 km/h
(c) 120 km/h
(d) 72 km/h
Answer: **(B)** | *Explanation:* Total distance to clear = 240 m + 160 m = 400 m.
Let speed of faster train = v1 m/s and slower train = v2 m/s.
1) Running in opposite direct...

Q23. In an election, 20,000 voters were registered. 80% voted, and 5% of the votes cast were invalid. The winner got 55% of the valid votes. By how many votes did he win?
(a) 1520
(b) 1600
(c) 1360
(d) 1440
Answer: **(A)** | *Explanation:* Total registered voters = 20,000.
Voters who voted = 80% of 20,000 = 16,000.
Invalid votes = 5% of 16,000 = 800.
Valid votes = 16,000 - 800 ...

Q24. The difference between compound interest (annual) and simple interest on a sum for 3 years at 10% p.a. is ₹310. Find the sum.
(a) ₹8,000
(b) ₹9,000
(c) ₹10,000
(d) ₹12,000
Answer: **(C)** | *Explanation:* Difference between CI and SI for 3 years at rate r%:

Formula: Difference = P × (r / 100)^2 × (3 + r / 100).
Given r = 10% and Difference = ₹3...

Q25. ₹12,000 is invested partly at 8% simple interest and the rest at 10% compounded annually. After 2 years the total interest is ₹2,270. How much was invested at simple interest?
(a) ₹4,000
(b) ₹5,000
(c) ₹6,000
(d) ₹7,000
Read the passage and answer Q26–30.
In software organisations, "technical debt" is often treated as a purely engineering concern, a messy residue of hurried decisions that developers will eventually tidy up. This framing is misleading. Debt, in the financial sense, is not inherently harmful; a company that borrows to seize a market opportunity may be acting shrewdly. The problem arises when the interest, meaning the extra effort every future change demands, is never acknowledged on the balance sheet. Teams then find that features which once took days now take weeks, yet managers, seeing only the delivered functionality, attribute the slowdown to declining developer productivity. Proponents of disciplined refactoring argue that a fixed share of every sprint should be reserved for repayment. Critics counter that such quotas are arbitrary and encourage cosmetic rewrites that leave real risks untouched. A more defensible approach, some suggest, is to make debt visible by tracking how often particular modules cause delays or defects, and to repay only where the interest is demonstrably high.
Answer: **(B)** | *Explanation:* Let sum invested at 8% SI = ₹x.
Then sum invested at 10% CI = ₹(12,000 - x).
Time = 2 years.
- Simple Interest at 8% for 2 years = (x × 8 × ...

SECTION 3: Verbal Ability (20 min)

Reading Comprehension Passage (Q26–Q30):

```
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the interest is demonstrably high.
```

Q26. The central idea of the passage is that:
(a) Technical debt is harmful and must be eliminated at once
(b) Technical debt is not necessarily bad, but its recurring costs must be made visible and repaid selectively
(c) Managers, not developers, are responsible for technical debt
(d) Fixed refactoring quotas are the best way to manage debt
Answer: **(B)** | *Explanation:* The passage argues that technical debt is analogous to financial debt: not inherently evil, but dangerous when its recurring carrying costs ...

Q27. According to the passage, managers blame developer productivity because they:
(a) do not trust developers
(b) see only the delivered features and not the hidden extra effort
(c) are unaware that software can be refactored
(d) have reserved too little time for sprints
Answer: **(B)** | *Explanation:* The passage states explicitly: 'yet managers, seeing only the delivered functionality, attribute the slowdown to declining developer product...

Q28. Which statement would critics of fixed repayment quotas most likely agree with?
(a) Repayment effort should target genuine risks rather than meet a set proportion
(b) Refactoring should be abandoned entirely
(c) Cosmetic rewrites are the most valuable form of repayment
(d) Sprints should be lengthened
Answer: **(A)** | *Explanation:* Critics argue that fixed quotas 'are arbitrary and encourage cosmetic rewrites that leave real risks untouched.' Hence, they would advocate ...

Q29. In the passage, "interest" most nearly refers to:
(a) money paid to lenders
(b) the recurring additional effort imposed on future changes
(c) the curiosity of developers
(d) the profit from a market opportunity

Answer: **(B)** | *Explanation:* The passage defines interest directly: 'the interest, meaning the extra effort every future change demands'....

Q30. Which of the following is NOT supported by the passage?
(a) Borrowing can sometimes be a shrewd decision
(b) A slowdown in delivery always indicates falling developer productivity
(c) Tracking defect-prone modules can help identify costly debt
(d) Debt is often left unrecorded in practice
Sentence Correction (Q31–35). Choose the best replacement for the bold part.
Answer: **(B)** | *Explanation:* The passage explicitly notes that a delivery slowdown is often caused by accumulated technical debt rather than declining developer producti...

Q31. Neither the manager nor the engineers was aware of the revised deadline.
(a) No improvement
(b) engineers were
(c) engineers has been
(d) engineer were
Answer: **(B)** | *Explanation:* With correlative conjunctions 'neither... nor', the verb agrees in number with the nearer subject. Here, the nearer subject is 'engineers' (...

Q32. Had the team would have tested the patch thoroughly, the outage could have been avoided.
(a) No improvement
(b) Had the team tested
(c) Has the team tested
(d) If the team would test
Answer: **(B)** | *Explanation:* Inverted third conditional structure uses 'Had + subject + past participle'. The conditional clause cannot contain 'would have'. Correct: 'H...

Q33. Hardly the server had restarted when the alerts began flooding in.
(a) No improvement
(b) had the server restarted when
(c) did the server restarted when
(d) the server
restarted than
Answer: **(B)** | *Explanation:* Negative adverbial 'Hardly' placed at the beginning of a clause triggers negative inversion: 'Hardly had + subject + past participle... when...

Q34. The number of applicants are increasing, but a large number of them lacks the required skills.
(a) are … lacks
(b) is … lack
(c) is … lacks
(d) are … lack
Answer: **(B)** | *Explanation:* 'The number of' denotes a singular collective quantity and takes a singular verb ('is increasing'). In contrast, 'a large number of' denotes...

Q35. Each of the interns, along with their mentor, were invited to present.
(a) No improvement
(b) was
(c) are

(d) have been
Para-jumbles (Q36–38). Arrange the sentences in the correct order.
Answer: **(B)** | *Explanation:* The true grammatical subject of the sentence is 'Each' (distributive pronoun, singular). Parenthetical prepositional phrases like 'along wit...

Q36. P: However, the prototype failed under real-world load.
Q: The engineers were confident that their design was flawless.
R: Post-mortem analysis revealed that no load testing had been performed.
S: Consequently, the launch was postponed by three months.
(a) QPRS
(b) QRPS
(c) PQRS
(d) QSPR
Answer: **(A)** | *Explanation:* Coherent narrative flow: Q introduces initial state ('The engineers were confident...'), P introduces the contrast ('However, the prototype ...

Q37. P: Rather, they encourage people to think about how the story might have unfolded differently.
Q: Counterfactual questions such as "What if the treaty had never been signed?" are popular with historians.
R: They do not aim to predict what would actually have happened.
S: Such speculation, when disciplined, can illuminate the forces that shaped real events.
(a) QRPS
(b) QPRS
(c) RQPS
(d) QRSP
Answer: **(A)** | *Explanation:* Logical progression: Q introduces the subject ('Counterfactual questions... are popular with historians'), R defines their negative boundary...

Q38. P: The first industrial robots were deployed on car assembly lines in the 1960s.
Q: Today, collaborative robots work safely alongside humans.
R: These early machines were caged off, since they could not sense human presence.
S: Advances in sensors and software have since removed this limitation.
(a) PRSQ
(b) PQRS
(c) PSRQ
(d) RPSQ
Vocabulary, Synonyms and Antonyms (Q39–45).
Answer: **(A)** | *Explanation:* Chronological progression: P establishes historical origin in the 1960s ('The first industrial robots were deployed...'), R describes their ...

Q39. The auditor's meticulous approach uncovered discrepancies others had missed. "Meticulous" means:
(a) hasty
(b) painstakingly careful
(c) aggressive
(d) casual
Answer: **(B)** | *Explanation:* 'Meticulous' means showing great attention to detail; very careful and precise. Hence, 'painstakingly careful'....

Q40. The contract was deliberately worded to obfuscate the penalty clauses. "Obfuscate" means:
(a) highlight
(b) simplify
(c) make obscure
(d) cancel
Answer: **(C)** | *Explanation:* 'Obfuscate' means to make obscure, unclear, or unintelligible....

Q41. Fill in the blank: The startup's success proved _____; within a year it had vanished.
(a) enduring
(b) ephemeral
(c) robust
(d) inevitable
Answer: **(B)** | *Explanation:* 'Ephemeral' means lasting for a very short time; transitory. Since the startup vanished within a year, its success was ephemeral....

Q42. Synonym of UBIQUITOUS:
(a) rare
(b) omnipresent
(c) ancient
(d) hidden
Answer: **(B)** | *Explanation:* 'Ubiquitous' means present, appearing, or found everywhere; omnipresent....

Q43. Synonym of RECALCITRANT:
(a) obedient
(b) obstinate
(c) cheerful
(d) timid
Answer: **(B)** | *Explanation:* 'Recalcitrant' means having an obstinately uncooperative attitude towards authority or discipline; obstinate....

Q44. Antonym of LOQUACIOUS:
(a) talkative
(b) taciturn
(c) eloquent
(d) boisterous
Answer: **(B)** | *Explanation:* 'Loquacious' means tending to talk a great deal; talkative. The antonym is 'taciturn' (reserved or uncommunicative in speech)....

Q45. Antonym of MAGNANIMOUS:
(a) generous
(b) noble
(c) petty
(d) tolerant
Answer: **(C)** | *Explanation:* 'Magnanimous' means generous or forgiving, especially towards a rival or someone less powerful. The antonym is 'petty' (narrow-minded, spite...

SECTION 4: Pseudo Code (10 min)

Q46. What is printed?

```
int f(int n) {
if (n <= 1) return n;
return f(n-1) + 2 * f(n-2);
}
print(f(6))
```

(a) 15
(b) 21
(c) 27
(d) 31

Answer: **(B)** | *Explanation:* Recursive function trace:
- f(0) = 0
- f(1) = 1
- f(2) = f(1) + 2·f(0) = 1 + 0 = 1
- f(3) = f(2) + 2·f(1) = 1 + 2(1) = 3
- f(4) = f(3) + 2·f...

Q47. What is printed?

```
x = 45; count = 0;
while (x != 0) {
x = x & (x - 1);
count++;
}
print(count ^ (45 >> 2))
```

(a) 4
(b) 11
(c) 15
(d) 7

Answer: **(C)** | *Explanation:* Brian Kernighan's bit-counting loop clears the lowest set bit in each iteration.
For x = 45 = 32 + 8 + 4 + 1 = 101101_2 (4 set bits):
Loop ex...

Q48. What is printed?

```
s = 0
for i = 1 to 5
for j = i to 5
if ((i + j) % 3 == 0) continue
if (j == 5) break
s = s + i * j
end for
end for
print(s)
```

(a) 38
(b) 42
(c) 46

(d) 50
Answer: **(C)** | *Explanation:* Nested loops simulation:
- i=1: j=1 (sum=2, add 1); j=2 ((1+2)%3==0, continue); j=3 (sum=4, add 3); j=4 (sum=5, add 4); j=5 ((1+5)%3==0, con...

Q49. A stack is used (operands are evaluated left to right). What is printed?
push(3); push(5); push(7);
x = pop();
push(x * 2);
push(pop() + pop());
print(pop() - pop())
(a) 14
(b) 16
(c) 19
(d) 22
Answer: **(B)** | *Explanation:* Stack execution trace (LIFO):
1) push(3) -> [3]
2) push(5) -> [3, 5]
3) push(7) -> [3, 5, 7]
4) x = pop() => x = 7; stack = [3, 5]
5) push(7...

Q50. What is the final content of arr? (indexing starts at 0; updates are in place)
arr = [4, 1, 3, 2]
for i = 0 to 3
arr[i] = arr[i] + arr[arr[i] % 4]
end for
(a) [8, 2, 5, 7]
(b) [8, 2, 5, 6]
(c) [8, 5, 5, 7]
(d) [5, 2, 5, 7]
Answer: **(A)** | *Explanation:* In-place array update with arr = [4, 1, 3, 2]:
- i = 0: arr[0] = arr[0] + arr[arr[0] % 4] = 4 + arr[4 % 4] = 4 + arr[0] = 4 + 4 = 8. arr = [...

SECTION 5: Numeric Puzzle Solving (10 min)

Q51. Matchstick. A 2 × 2 grid of squares is made using 12 matchsticks (giving 4 small squares and 1 large square). What is the minimum number of sticks to remove so that no square of any size remains?
(a) 2
(b) 3
(c) 4
(d) 5
Answer: **(B)** | *Explanation:* In a 2 × 2 matchstick grid (12 sticks total, forming four 1×1 squares and one 2×2 square):
To eliminate all 5 squares with the minimum numbe...

Q52. Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.

Wheel 1: T = 3, R = 5, B = 4, L = 7, Centre = 14
Wheel 2: T = 2, R = 6, B = 5, L = 3, Centre = 7
Wheel 3: T = 4, R = 3, B = 6, L = 9, Centre = ?
(a) 21
(b) 27
(c) 30
(d) 33
Answer: **(C)** | *Explanation:* Determining the circular wheel formula:
- Wheel 1: T=3, R=5, B=4, L=7, Centre=14.
Formula: Centre = (T × B) + L - R = (3 × 4) + 7 - 5 = 12...

Q53. Matrix logic. Find the missing number.
3
4
25
5
12
169
6
8
?
(a) 64
(b) 100
(c) 196
(d) 144
Answer: **(B)** | *Explanation:* Matrix logic pattern:
Row 1: 3, 4, 25 => $3^2 + 4^2 = 9 + 16 = 25$.
Row 2: 5, 12, 169 => $5^2 + 12^2 = 25 + 144 = 169$.
Row 3: 6, 8, ? => $6^2 + 8^2 = ...$

Q54. Pattern. Find the next term: 2, 12, 36, 80, 150, ?
(a) 216
(b) 252
(c) 240
(d) 294
Answer: **(B)** | *Explanation:* Sequence: 2, 12, 36, 80, 150, ?
Pattern: $n^3 + n^2$ for n = 1, 2, 3, 4, 5, ...
- n = 1: $1^3 + 1^2 = 1 + 1 = 2$
- n = 2: $2^3 + 2^2 = 8 + 4 = 12$
- n =...

SECTION 6: English Grammar (10 min)

Q55. Neither of the two proposals, despite several revisions, _____ approved.
(a) have been
(b) has been
(c) were

(d) are
Answer: **(B)** | *Explanation:* 'Neither of' followed by a plural noun phrase ('the two proposals') takes a singular verb. The intervening parenthetical phrase ('despite se...

Q56. By the time the auditors arrive next month, we _____ the migration.
(a) completed
(b) will complete
(c) will have completed
(d) have completed
Answer: **(C)** | *Explanation:* The time marker 'By the time + simple present clause' ('the auditors arrive next month') establishes a deadline in the future before which a...

Q57. The manager insisted _____ reviewing every line of code, despite being averse _____ delay.
(a) on, to
(b) in, from
(c) upon, with
(d) at, to
Answer: **(A)** | *Explanation:* Fixed prepositional collocations: 'insisted on/upon' and 'averse to'. Therefore, 'on, to'....

Q58. Error spotting. Find the part with the error: "Not only did the CEO apologise (A) / but he also (B) / promised to (C) / compensated the customers. (D)"
(a) A
(b) B
(c) C
(d) D
Answer: **(D)** | *Explanation:* Infinitive structure: 'promised to + base verb'. 'Compensated' is past tense; it must be bare infinitive 'compensate'. The error is in part ...

Q59. She is ___ European who works as ___ engineer at ___ MNC.
(a) a, an, an
(b) an, an, a
(c) a, an, a
(d) a, a, an
Answer: **(A)** | *Explanation:* Article rules based on phonetic pronunciation:
1) 'European' begins with consonant glide /j/ ('yoo-roh-pee-an') -> takes indefinite article ...

SECTION 7: English Writing (10 min)

Q60. Write an email of 120–150 words.
You are a System Engineer in a project team. A production deployment scheduled for tonight has to be postponed by 48 hours because a critical defect was found during final testing. Write an email to the client project manager that informs them of the delay, explains the reason, states the corrective action and the revised timeline, and apologises for any inconvenience.
Evaluation rubric (10 marks): Subject line and format (2) | Clarity and completeness of content (3) | Professional tone and courtesy (2) | Grammar, spelling and vocabulary (2) | Word limit adherence (1)
Answer: **(N/A)** | *Explanation:* Evaluation Rubric (10 Marks): Subject line and format (2) | Content clarity and completeness (3) | Professional tone and courtesy (2) | Gram...

Mock Exam Paper 02: Infosys System Engineer ITM Mock Paper 2

Source File: `Infosys_SE_Mock_Paper_2.pdf`
Total Questions: 60 Questions | **Total Marks:** 69 Marks (59 MCQs + 10 Marks Writing)

SECTION 1: Reasoning Ability (25 min)

Q01. Statements: All servers are machines. Some machines are laptops. No laptop is portable.
Conclusions: I. Some machines are not portable. II. Some servers are laptops. III. No server is portable.
(a) Only I follows
(b) I and III follow
(c) I and II follow
(d) None follows
Answer: **(A)** | *Explanation:* Some machines are laptops, and no laptop is portable, so those machines are not portable — I follows.
Nothing links servers specifically to...

Q02. Statements: All circles are squares. No square is a triangle. Some triangles are rectangles.
Conclusions: I. No circle is a triangle. II. Some rectangles are not squares. III. Some squares are not triangles.
(a) I and III only
(b) I only
(c) I, II and III
(d) II and III only
Answer: **(C)** | *Explanation:* Circles
squares, and squares have no overlap with triangles, so circles have none either (I). Some
$\subseteq$
triangles are rectangles, and since ...

Q03. Data Sufficiency. Is the integer n divisible by 6?
I. n is divisible by 4. II. n is divisible by 9.
(a) I alone is sufficient
(b) II alone is sufficient
(c) Both together are sufficient, neither alone
(d) Even both
together are not sufficient
Answer: **(C)** | *Explanation:* I alone fails (n = 4 is not divisible by 6). II alone fails (n = 9 is not divisible by 6). Together, a number
divisible by both 4 and 9 is ...

Q04. Data Sufficiency. What is the cost price of an article?
I. It was sold at a profit of 20%. II. If sold at ₹150 more, the profit would have been 32%.
(a) I alone is sufficient
(b) II alone is sufficient
(c) Both together are sufficient, neither alone
(d) Even both
together are not sufficient
Answer: **(B)** | *Explanation:* II alone: 12% of CP = ₹150, so CP = ₹1,250 — sufficient. I alone gives only a

percentage, not an absolute
value....

Q05. In a code, every letter is shifted 3 places forward in the alphabet (wrapping Z→C), so FLOWER is written as IORZHU. Using the same code, how is GARDEN written?
(a) JDUGHQ
(b) JDUFHQ
(c) JCUGHQ
(d) JDVGHQ
Answer: **(A)** | *Explanation:* Applying the same +3 shift: G→J, A→D, R→U, D→G, E→H, N→Q, giving JDUGHQ....

Q06. In a code, "data flows fast" = "zu ka mo", "fast network grows" = "ka ta ri", "network data secure" = "zu ta ve". What is the code for "grows secure data"?
(a) ri ve zu
(b) ta ve zu
(c) ri ve ka
(d) ka ve zu
Answer: **(A)** | *Explanation:* "fast" is common to statements 1 and 2 → ka. "data" is common to 1 and 3 → zu. "network" is
common to 2 and 3 → ta. By elimination: flows =...

Q07. A cube is painted red on one pair of opposite faces, blue on another pair, and green on the last pair. It is then cut into 27 identical small cubes. How many small cubes have exactly one face painted?
(a) 6
(b) 8
(c) 9
(d) 12
Answer: **(A)** | *Explanation:* In a 3×3×3 cube, only the centre cube of each face touches just one painted face — 1 per face × 6 faces
= 6, regardless of the colour schem...

Q08. A rectangle is divided into a 3 × 3 grid of smaller cells by 2 evenly spaced vertical lines and 2 evenly spaced horizontal lines drawn inside it. How many rectangles (of any size) are there in the whole figure?
(a) 30
(b) 36
(c) 27
(d) 24
Answer: **(B)** | *Explanation:* With 2 interior + 2 boundary lines in each direction, there are 4 vertical and 4 horizontal lines.
Rectangles = C(4,2) × C(4,2) = 6 × 6 = 3...

Q09. Pointing to a photograph, a man says, "She is the daughter of my grandfather's only son." How is the woman in the photograph related to the man?
(a) Sister
(b) Daughter
(c) Niece
(d) Mother

Answer: **(A)** | *Explanation:* The grandfather's only son is the man's father. The daughter of the man's father is his sister....

Q10. If A ÷ B means A is the daughter of B, A + B means A is the son of B, and A − B means A is the wife of B, which expression correctly shows that M is the granddaughter of N?
(a) M ÷ P + N
(b) M + P ÷ N
(c) M − P ÷ N
(d) M ÷ P − N
Answer: **(A)** | *Explanation:* M ÷ P: M is daughter of P. P + N: P is son of N. So M is P's daughter, and P is N's son — M is N's
granddaughter....

Q11. Priya starts at a point, faces south and walks 8 m. She turns left and walks 5 m, turns left again and walks 8 m, then turns right and walks 3 m. How far is she from the starting point, and in which direction?
(a) 8 m East
(b) 8 m West
(c) 2 m East
(d) 2 m West
Answer: **(A)** | *Explanation:* Tracking positions from (0,0): south 8 → (0,−8); left turn faces east, walk 5 → (5,−8); left turn faces
north, walk 8 → (5,0); right turn f...

Q12. A man facing South turns 45° anticlockwise, then 90° clockwise, then 225° anticlockwise. Which direction is he facing now?
(a) North
(b) South-East
(c) South-West
(d) West
Answer: **(A)** | *Explanation:* Using bearings (clockwise from North): South = 180°. −45° (anticlockwise) = 135°; +90° (clockwise) =
225°; −225° (anticlockwise) = 0°, whic...

Q13. Find the next term: 2, 6, 15, 31, 56, ?
(a) 92
(b) 88
(c) 90
(d) 84
Answer: **(A)** | *Explanation:* Differences are 4, 9, 16, 25 (perfect squares), so the next difference is 36: 56 + 36 = 92....

Q14. Find the next term: 5, 11, 23, 47, 95, ?
(a) 189
(b) 191
(c) 193
(d) 190
Answer: **(B)** | *Explanation:* Each term = previous × 2 + 1: 95 × 2 + 1 = 191....

Q15. Find the next letter-pair: CE, FI, IM, LQ, ?
(a) OT
(b) OU
(c) NT
(d) PU
Answer: **(B)** | *Explanation:* First letters C, F, I, L rise by 3 each time → next is O. Second letters E, I, M, Q rise by 4 each time →
next is U. Answer: OU.
Page 8 of ...

SECTION 2: Technical Ability (Quantitative) (35 min)

Q16. In how many ways can the letters of the word MANAGEMENT be arranged so that all the vowels come together?
(a) 7560
(b) 15120
(c) 3780
(d) 30240
Answer: **(A)** | *Explanation:* MANAGEMENT has M2, A2, N2, G1, E2, T1 (10 letters). Vowels A,A,E,E form one block: internal
arrangements = 4!/(2!2!) = 6. Treating the bloc...

Q17. From a group of 6 men and 5 women, a team of 4 is to be selected such that it contains at most 2 women. In how many ways can this be done?
(a) 265
(b) 295
(c) 310
(d) 325
Answer: **(A)** | *Explanation:* 0 women: C(6,4)=15. 1 woman: C(6,3)×C(5,1)=100. 2 women: C(6,2)×C(5,2)=150. Total = 15+100+150
= 265....

Q18. Three coins are tossed together, and it is known that at least one head appeared. What is the probability that exactly two heads appeared?
(a) 3/7
(b) 1/2
(c) 3/8
(d) 2/7
Answer: **(A)** | *Explanation:* Excluding TTT leaves 7 equally likely outcomes. Exactly two heads: HHT, HTH, THH = 3 outcomes.
Probability = 3/7....

Q19. A box contains 6 defective and 14 non-defective bulbs. Two bulbs are drawn at random without replacement. What is the probability that at least one is defective?
(a) 99/190
(b) 91/190
(c) 33/95
(d) 29/95

Answer: **(A)** | *Explanation:* P(no defective) = (14×13)/(20×19) = 91/190. P(at least one defective) = 1 − 91/190 = 99/190....

Q20. A trader marks his goods at a price that would give him a 25% profit, but he actually gives a discount of 12% on the marked price and also uses a weight that is 10% less than it should be. Find his actual profit percentage.
(a) 20%
(b) 22.2%
(c) 25%
(d) 18%
Answer: **(B)** | *Explanation:* Take CP = 100 per unit weight, so marked price = 125 (25% intended profit). After a 12% discount, SP
= 125 × 0.88 = 110. But he delivers on...

Q21. A and B together can complete a task in 15 days. They worked together for 5 days, after which A left and B completed the remaining work alone in 20 more days. In how many days can A alone complete the task?
(a) 20 days
(b) 24 days
(c) 25 days
(d) 30 days
Answer: **(D)** | *Explanation:* In 5 days together, they finish 5/15 = 1/3. The remaining 2/3 is done by B alone in 20 days, so B's rate
= (2/3)/20 = 1/30/day (B alone tak...

Q22. A boat's speed in still water is 15 km/h. It travels 60 km downstream and returns upstream to the starting point, taking a total of 9 hours. Find the speed of the stream.
(a) 4 km/h
(b) 5 km/h
(c) 6 km/h
(d) 3 km/h
Answer: **(B)** | *Explanation:* 60/(15+s) + 60/(15−s) = 9. Simplifying: 1800 = 9(225 − s^2), so s^2 = 25, s = 5 km/h....

Q23. A town's population increases by 10% in the first year and decreases by 8% in the second year. If the population at the end of two years is 27,324, what was the population at the start?
(a) 27,000
(b) 26,500
(c) 26,000
(d) 27,500
Answer: **(A)** | *Explanation:* P × 1.10 × 0.92 = 27,324, so P × 1.012 = 27,324, giving P = 27,000....

Q24. A sum of money amounts to ₹13,000 after 3 years and to ₹15,000 after 5 years at the same rate of simple interest. Find the rate of interest per annum.
(a) 8%
(b) 10%
(c) 12%
(d) 9%
Answer: **(B)** | *Explanation:* Interest for 2 years (year 5 − year 3) = 15,000 − 13,000 = 2,000, so 1 year's interest =

1,000. Principal =
13,000 − 3×1,000 = 10,000. Rate...

Q25. ₹16,000 is invested for 2 years at compound interest, compounded annually, and yields the same interest as ₹18,000 invested for 2 years at 10% simple interest. Find the rate of compound interest (approximately).
(a) 9.5%
(b) 10.5%
(c) 11.5%
(d) 12.5%
Read the passage and answer Q26–30.
Automation in enterprise software is frequently sold on the promise of eliminating human error, yet the more interesting effect is often a redistribution of error rather than its removal. When a system automates data entry, the mistakes that once occurred at the point of typing largely disappear, but new failure modes emerge upstream, in the design of the rules the automation follows, and downstream, in the operators' declining vigilance once they trust the system implicitly. A well-documented pattern in aviation and, increasingly, in IT operations is that highly reliable automation breeds complacency: engineers stop cross-checking outputs precisely because the system rarely errs, so on the rare occasion it does, the mistake propagates further before anyone notices. This does not argue against automation; it argues for designing monitoring and escalation processes that assume the automation will occasionally fail silently, rather than processes that assume a human will always catch what the machine misses.
Answer: **(B)** | *Explanation:* SI on ₹18,000 at 10% for 2 years = ₹3,600. Setting CI on ₹16,000 equal:
$16{,}000\times((1+r)^2 - 1) = 3{,}600$
gives $(1+r)^2 = 1.225$, so r ≈ 10.5%.
Sec...

SECTION 3: Verbal Ability (20 min)

Reading Comprehension Passage (Q26–Q30):

```
Read the passage and answer Q26–30.
Automation in enterprise software is frequently sold on the promise of eliminating human error,
yet the
more interesting effect is often a redistribution of error rather than its removal. When a system
automates data entry, the mistakes that once occurred at the point of typing largely disappear,
but new
failure modes emerge upstream, in the design of the rules the automation follows, and downstream,
in
the operators' declining vigilance once they trust the system implicitly. A well-documented
pattern in
aviation and, increasingly, in IT operations is that highly reliable automation breeds
complacency:
engineers stop cross-checking outputs precisely because the system rarely errs, so on the rare
occasion it
does, the mistake propagates further before anyone notices. This does not argue against
automation; it
argues for designing monitoring and escalation processes that assume the automation will
occasionally
fail silently, rather than processes that assume a human will always catch what the machine
misses.
```

Q26. The passage primarily argues that:
(a) Automation should be avoided in critical systems
(b) Automation shifts the location of errors rather than eliminating them, so monitoring must assume silent failure
(c) Human operators are always more reliable than automated systems
(d) Aviation and IT face identical automation challenges
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 2, option (B) is directly supported and established by the author's primary thesis ...

Q27. According to the passage, complacency arises because:
(a) operators are poorly trained
(b) the automation is rarely wrong, so vigilance declines
(c) the rules are too complex to follow
(d) engineers do not trust the system
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 2, option (B) is directly supported and established by the author's primary thesis ...

Q28. The phrase "fail silently" in the passage most nearly means:
(a) the system stops working with an alarm
(b) the system errs without producing any visible warning
(c) the system asks for human confirmation
(d) the system shuts down automatically
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 2, option (B) is directly supported and established by the author's primary thesis ...

Q29. Which of these would the author most likely recommend?
(a) Removing automation from enterprise systems entirely
(b) Designing escalation processes that do not rely solely on human vigilance
(c) Increasing the frequency of manual data entry
(d) Trusting automated systems completely once proven reliable
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 2, option (B) is directly supported and established by the author's primary thesis ...

Q30. Which of the following is NOT stated or implied in the passage?
(a) New failure modes can appear in the design of automation rules
(b) Reliable automation can reduce the vigilance of human operators
(c) The passage argues automation should never be trusted
(d) Errors caught late can propagate further
Sentence Correction (Q31–35).
Answer: **(C)** | *Explanation:* According to the reading comprehension passage for Paper 2, option (C) is directly supported and established by the author's primary thesis ...

Q31. Each of the servers in the cluster have its own dedicated backup.
(a) No improvement
(b) has
(c) having
(d) had
Answer: **(B)** | *Explanation:* "Each" is singular, so the verb must be "has"....

Q32. Scarcely had the deployment finished than the monitoring alerts started.
(a) No improvement
(b) Scarcely had the deployment finished when
(c) Scarcely did the deployment finish
than
(d) Scarcely the deployment had finished when
Answer: **(B)** | *Explanation:* "Scarcely...when" is the correct pairing, not "than"....

Q33. The report, along with its supporting data sheets, were submitted before the deadline.
(a) No improvement
(b) was submitted
(c) have been submitted
(d) are submitted
Answer: **(B)** | *Explanation:* "The report" is the true (singular) subject; "along with..." does not pluralize it, so "was submitted"....

Q34. If I was you, I would escalate this incident immediately.
(a) No improvement
(b) If I were you
(c) If I am you
(d) If I had been you
Answer: **(B)** | *Explanation:* Hypothetical conditions use the subjunctive "were" after "if", regardless of subject....

Q35. Not only the developer but also the testers was responsible for the missed defect.
(a) No improvement
(b) were responsible
(c) is responsible
(d) had been responsible
Para-jumbles (Q36–38).
Answer: **(B)** | *Explanation:* With "not only...but also" joining two subjects, the verb agrees with the nearer one (testers, plural),
so "were responsible"....

Q36. P: Yet within weeks, unexplained crashes began appearing in production.
Q: The new caching layer initially seemed to improve response times dramatically.
R: Investigation traced the crashes to a race condition the load tests had never triggered.
S: The team subsequently rewrote the load-testing suite to include concurrent write scenarios.
(a) QPRS
(b) QRPS
(c) PQRS
(d) QSPR
Answer: **(A)** | *Explanation:* Sentence sequence: Q→P→R→S.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q37. P: They are, instead, thought experiments that isolate one variable at a time.
Q: Economic models are often criticised for their unrealistic assumptions.
R: This deliberate simplification is what makes them useful, not despite the unrealism but because of it.
S: Critics forget that the models were never meant to be literal descriptions of the economy.

(a) QSPR
(b) QPSR
(c) QSRP
(d) SQPR
Answer: **(A)** | *Explanation:* Sentence sequence: Q→S→P→R.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q38. P: Consequently, the interface today handles ten times the daily traffic it was designed for.
Q: The original API was built to serve a handful of internal tools.
R: Over time, external partners began integrating with it as well.
S: No one anticipated that this informal adoption would continue for years.
(a) QRSP
(b) QSRP
(c) QRPS
(d) RQSP
Vocabulary, Synonyms and Antonyms (Q39–45).
Answer: **(A)** | *Explanation:* Sentence sequence: Q→R→S→P.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q39. The consultant's cursory review missed several critical vulnerabilities. "Cursory" means:
(a) thorough
(b) hasty and superficial
(c) expensive
(d) delayed
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q40. The new policy was met with tacit approval from the board. "Tacit" means:
(a) loudly announced
(b) implied without being stated
(c) formally written
(d) rejected
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q41. Fill in the blank: Despite the outage, the support team remained _____, calmly working through the incident.
(a) frantic
(b) unflappable
(c) indifferent
(d) negligent
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q42. Synonym of PRAGMATIC:
(a) idealistic
(b) practical
(c) theoretical

(d) emotional
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q43. Synonym of VOLATILE:
(a) stable
(b) unpredictable
(c) calm
(d) gradual
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q44. Antonym of CONCISE:
(a) brief
(b) verbose
(c) clear
(d) precise
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q45. Antonym of DILIGENT:
(a) hardworking
(b) careful
(c) lazy
(d) thorough
Answer: **(C)** | *Explanation:* Section 4: Pseudo Code...

SECTION 4: Pseudo Code (10 min)

Q46. What is printed?
```
int f(int n) {
if (n == 0) return 1;
if (n == 1) return 2;
return f(n-1) + f(n-2) * 2;
}
print(f(5))
```
(a) 26
(b) 30
(c) 32
(d) 34
Answer: **(C)** | *Explanation:* 32. f(0)=1, f(1)=2, f(2)=2+2=4, f(3)=4+4=8, f(4)=8+8=16, f(5)=16+16=32....

Q47. What is printed? (x and y are unsigned integers)
```
x = 58; y = 0;
while (x > 0) {
y = (y << 1) | (x & 1);
x = x >> 1;
}
```

print(y)
(a) 23
(b) 29
(c) 37
(d) 41
Answer: **(A)** | *Explanation:* 23. The loop peels off bits of x from the LSB and rebuilds y by shifting left each time, which reverses
the bit order of x within its signi...

Q48. What is printed?
s = 0
for i = 1 to 4
for j = 1 to 4
if (i == j) continue
if (i + j > 6) break
s = s + (i * j)
end for
end for
print(s)
(a) 40
(b) 44
(c) 46
(d) 48
Answer: **(C)** | *Explanation:* 46. i=1: skips j=1 (i==j); adds 1×2+1×3+1×4 = 9. i=2: skips j=2; adds 2×1+2×3+2×4 = 16 (running 25;
note j=4 gives i+j=6, which is not >6, ...

Q49. A queue is used (FIFO). What is printed?
enqueue(2); enqueue(4); enqueue(6);
a = dequeue();
enqueue(a + 10);
b = dequeue();
c = dequeue();
print(b + c)
(a) 10
(b) 16
(c) 18
(d) 22
Answer: **(A)** | *Explanation:* 10. Queue [2,4,6] → dequeue 2 (a) → [4,6] → enqueue 12 → [4,6,12] → dequeue 4 (b) → dequeue 6
(c) → [12]. b + c = 4 + 6 = 10....

Q50. What is the final content of arr? (0-indexed; each statement uses the array's state at that point)
arr = [1, 2, 3, 4, 5]
temp = arr[0]
arr[0] = arr[4]
arr[4] = arr[2]

arr[2] = arr[1]
arr[1] = temp
(a) [5, 1, 2, 4, 3]
(b) [5, 1, 3, 4, 2]
(c) [5, 2, 1, 4, 3]
(d) [5, 1, 2, 4, 2]
Answer: **(A)** | *Explanation:* temp=1. arr[0]=arr[4]=5 → [5,2,3,4,5]. arr[4]=arr[2]=3 → [5,2,3,4,3]. arr[2]=arr[1]=2 → [5,2,2,4,3].
arr[1]=temp=1 → final [5,1,2,4,3].
Pag...

SECTION 5: Numeric Puzzle Solving (10 min)

Q51. Matchstick. Five identical unit squares are arranged in a row sharing sides, using 16 matchsticks in total. What is the minimum number of sticks that must be removed so that exactly 3 complete unit squares remain?
(a) 1
(b) 2
(c) 3
(d) 4
Answer: **(A)** | *Explanation:* 1. The stick shared between the 2nd and 3rd squares is a side of both. Removing that single stick
breaks both squares at once, leaving the ...

Q52. Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 6, R = 2, B = 4, L = 3, Centre = 30
Wheel 2: T = 5, R = 3, B = 8, L = 2, Centre = 46
Wheel 3: T = 7, R = 4, B = 3, L = 6, Centre = ?
(a) 39
(b) 41
(c) 45
(d) 49
Answer: **(C)** | *Explanation:* 45. Rule: Centre = (T × B) + (R × L). Wheel 1: (6×4)+(2×3) = 24+6 = 30
. Wheel 2: (5×8)+(3×2) = 40+6 =
✓

46
. Wheel 3: (7×3)+(4×6) = 21+...

Q53. Matrix logic. Find the missing number.
2
3
11
3
4
19
4

5
?
(a) 27
(b) 29
(c) 31
(d) 33
Answer: **(B)** | *Explanation:* 29. Rule: third column = (a×b) + a + b. Row 1: (2×3)+2+3 = 11
. Row 2: (3×4)+3+4 = 19
. Row 3:
✓
✓
(4×5)+4+5 = 20+9 = 29....

Q54. Pattern. Find the next term: 1, 4, 10, 20, 35, ?
(a) 52
(b) 56
(c) 60
(d) 49
Answer: **(B)** | *Explanation:* 56. These are tetrahedral numbers, C(n+2,3): 1, 4, 10, 20, 35, and next is C(8,3) = 56.
Section 6: Grammar...

SECTION 6: English Grammar (10 min)

Q55. Each of the following options _____ its own trade-offs, so the team must weigh them carefully.
(a) have
(b) has
(c) having
(d) were having
Answer: **(B)** | *Explanation:* "Each" is singular → has....

Q56. By next Friday, the QA team _____ all the regression test cases for this release.
(a) finished
(b) will finish
(c) will have finished
(d) finishes
Answer: **(C)** | *Explanation:* Future perfect ("will have finished") for an action complete before a future point....

Q57. The architect is known _____ his attention to detail but is often impatient _____ incomplete requirements.
(a) for, with
(b) of, by
(c) for, of
(d) by, for
Answer: **(A)** | *Explanation:* known for, impatient with....

Q58. Error spotting. "The manager, together with his team members, (A) / are reviewing the proposal (B) / that was submitted (C) / earlier this week. (D)"
(a) A

(b) B
(c) C
(d) D
Answer: **(B)** | *Explanation:* The true subject is "The manager" (singular); "together with his team members" doesn't pluralize it,
so it should be "is reviewing" — the e...

Q59. He bought ___ umbrella, ___ one-way ticket, and ___ useful travel guide before ___ trip.
(a) an, a, a, the
(b) a, a, an, the
(c) an, a, a, a
(d) a, an, a, the
Answer: **(A)** | *Explanation:* an umbrella (vowel sound), a one-way ticket ("w" sound), a useful travel guide ("y" sound), the trip
(specific, already referenced).
Sectio...

SECTION 7: English Writing (10 min)

Q60. Write an email of 120–150 words.
You are a System Engineer supporting a client's application in production. You have identified that a recurring performance issue is caused by an outdated server configuration, and a fix requires a scheduled maintenance window that will cause 30 minutes of downtime. Write an email to the client's IT coordinator explaining the issue, proposing a suitable maintenance window, describing the expected impact, and requesting their approval to proceed.
Evaluation rubric (10 marks): Subject line and format (2) | Clarity and completeness of content (3) | Professional tone and courtesy (2) | Grammar, spelling and vocabulary (2) | Word limit adherence (1)
Answer: **(N/A)** | *Explanation:* Evaluation Rubric (10 Marks): Subject line & format (2) | Content clarity (3) | Professional tone (2) | Grammar & vocabulary (2) | Word limi...

Mock Exam Paper 03: Infosys System Engineer ITM Mock Paper 3

Source File: `Infosys_SE_Mock_Paper_3.pdf`
Total Questions: 60 Questions | **Total Marks:** 69 Marks (59 MCQs + 10 Marks Writing)

SECTION 1: Reasoning Ability (25 min)

Q01. Statements: All P are Q. All Q are R. Some R are S. No S is T.
Conclusions: I. All P are R. II. Some Q are S. III. No P is T. IV. Some R are not T.
(a) Only I follows
(b) I and IV follow
(c) I, II and IV follow
(d) All follow
Answer: **(B)** | *Explanation:* Since P⊆Q⊆R, "All P are R" (I) follows directly. Some R are S, and no S is T, so those particular R-
elements are not T — "Some R are not T"...

Q02. Statements: All coins are metals. No metal is wood. Some woods are furniture.
Conclusions: I. No coin is wood. II. Some furniture is not metal.
(a) Only I follows
(b) Only II follows
(c) Both I and II follow
(d) Neither follows
Answer: **(C)** | *Explanation:* Coins⊆metals, and metals∩wood=∅, so coins∩wood=∅ — I follows. Some woods are furniture, and
since no metal is wood, those wood-furniture it...

Q03. Data Sufficiency. What is the two-digit number N?
I. The sum of its digits is 9.
II. The number obtained by reversing its digits is 27 less than N.
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Each statement alone is sufficient
(e) Even both statements together are not sufficient
Answer: **(C)** | *Explanation:* II alone: N − reversed = 9(tens digit − units digit) = 27, so tens − units = 3 — multiple numbers fit (41,
52, 63, 74, 85, 96), insufficien...

Q04. Data Sufficiency. Is the integer x even?
I. x/2 is an integer.
II. x^2 is even.
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Each statement alone is sufficient
(e) Even both statements together are not sufficient
Answer: **(D)** | *Explanation:* I alone: x/2 being an integer directly means x is even — sufficient. II alone: x^2 even is only possible if x
itself is even (an odd number ...

Q05. In a certain code, TRAIN is written as 23145, GAME is written as 6178, and MEAT is written as 7812. What is the code for GATE?
(a) 6128
(b) 6218
(c) 6182
(d) 8126
Answer: **(A)** | *Explanation:* From TRAIN=23145: T=2,R=3,A=1,I=4,N=5. From GAME=6178: G=6,A=1,M=7,E=8. From MEAT=7812:
confirms M=7,E=8,A=1,T=2. So GATE = G(6)A(1)T(2)E(8...

Q06. In a code: "bright light shines" = "mo ta pu", "shines through night" = "pu ri ke", "night bright stars" = "ke
mo lo", "stars through galaxy" = "lo ri zu". What is the code for "night light galaxy"?
(a) ke ta zu

(b) ke ta ri
(c) mo ta zu
(d) ke lo zu
Answer: **(A)** | *Explanation:* Comparing statement 1 & 2, the common word "shines" gets the common code "pu". Comparing 2 &
3, "night" gets the common code "ke". Comparin...

Q07. A cuboid measuring 5 × 4 × 3 units is painted on all its outer surfaces and then cut into unit cubes. How many unit cubes have exactly 2 painted faces?
(a) 24
(b) 22
(c) 20
(d) 26
Answer: **(A)** | *Explanation:* For a cuboid a×b×c, the number of unit cubes with exactly 2 painted faces = 4[(a−2)+(b−2)+(c−2)]. With
a=5,b=4,c=3: 4[3+2+1] = 4×6 = 24....

Q08. How many triangles are there in total in a complete five-pointed star (pentagram) figure?
(a) 5
(b) 10
(c) 35
(d) 15
Answer: **(C)** | *Explanation:* A complete five-pointed star (pentagram) figure contains 35 triangles in total (5 small tip triangles, 5
medium, 5 larger, and combinations...

Q09. Introducing a man, a woman says, "His wife is the only daughter-in-law of my father." How is the woman related to the man's wife?
(a) Sister-in-law
(b) Sister
(c) Mother
(d) Aunt
Answer: **(A)** | *Explanation:* The man's wife is the only daughter-in-law of the woman's father, meaning the woman's father has
only one son — the man himself. So the man...

Q10. If A % B means A is the mother of B, and A * B means A is the father of B, which expression shows that C is the paternal grandmother of D?
(a) C % E * D
(b) C * E % D
(c) E % C * D
(d) D % E * C
Answer: **(A)** | *Explanation:* C % E: C is mother of E. E * D: E is father of D. So C is the mother of D's father, i.e., C is D's paternal
grandmother....

Q11. A man walks 12 m North, then 9 m East, then 8 m South, then 6 m West. How far is he from the starting point, and in which general direction?

(a) 5 m, North-East
(b) 5 m, North-West
(c) 7 m, North-East
(d) 5 m, South-East
Answer: **(A)** | *Explanation:* Net North-South displacement = 12−8 = 4 m North; net East-West = 9−6 = 3 m East. Distance =
$\sqrt{(3^2+4^2)} = \sqrt{25} = 5$ m, in the general North-East...

Q12. A cyclist starts facing East. He turns 135° clockwise, then 90° anticlockwise, then 180° clockwise. Which direction is he facing now?
(a) North-West
(b) South-East
(c) South-West
(d) North-East
Answer: **(A)** | *Explanation:* Using bearings (clockwise from North, East=90°): 90°+135°=225°(SW); 225° −90°=135°(SE);
135°+180°=315°, which is North-West....

Q13. Find the next term: 7, 26, 63, 124, 215, ?
(a) 342
(b) 344
(c) 341
(d) 343
Answer: **(A)** | *Explanation:* Each term is n^3-1 for n=2,3,4,5,6: 7,26,63,124,215; next (n=7): 343−1 = 342....

Q14. Find the next term: 4, 5, 9, 18, 34, ?
(a) 55
(b) 59
(c) 63
(d) 57
Answer: **(B)** | *Explanation:* Differences are 1,4,9,16 (perfect squares); next difference is 25: 34+25 = 59....

Q15. Find the next term: B, D, G, K, P, ?
(a) V
(b) U
(c) T
(d) W
Answer: **(A)** | *Explanation:* Differences between consecutive letters (by position) increase by 1 each time: +2,+3,+4,+5, next +6:
P(16)+6 = 22 = V.

...

SECTION 2: Technical Ability (Quantitative) (35 min)

Q16. In how many ways can 8 different books be distributed among 3 students so that one student gets 4 books and the other two get 2 books each?
(a) 1260

(b) 2520
(c) 630
(d) 420
Answer: **(A)** | *Explanation:* Choose which of the 3 students gets 4 books: 3 ways. Choose that student's 4 books: C(8,4)=70. Split
the remaining 4 books between the othe...

Q17. A committee of 6 is to be formed from 7 men and 6 women such that the committee has more men than women. In how many ways can this be done?
(a) 658
(b) 686
(c) 700
(d) 623
Answer: **(A)** | *Explanation:* Valid (men,women) splits with men>women, total=6: (6,0): C(7,6)×C(6,0)=7. (5,1): C(7,5)×C(6,1)=21×6=126. (4,2): C(7,4)×C(6,2)=35×15=525. To...

Q18. Bag A has 4 red and 6 black balls; Bag B has 7 red and 3 black balls. A bag is chosen at random and one ball is drawn from it. Given that the ball drawn is red, what is the probability that it came from Bag A?
(a) 4/11
(b) 7/11
(c) 2/5
(d) 1/2
Answer: **(A)** | *Explanation:* P(Red|A)=4/10, P(Red|B)=7/10, each bag chosen with probability 1/2. P(Red) = ½×0.4 + ½×0.7 = 0.55.
By Bayes' theorem, P(A|Red) = (½×0.4)/0....

Q19. A drawer has 5 red pens, 4 blue pens and 3 black pens. Four pens are drawn at random. What is the probability that exactly 2 are red and 2 are blue?
(a) 4/33
(b) 3/33
(c) 5/33
(d) 4/29

Sokware effort estimation has resisted four decades of methodological refinement, and the reason is not, as oken supposed, a lack of adequate techniques. Function point analysis, story point velocity tracking and Monte Carlo simulation are all, in controlled retrospective studies, reasonably well calibrated. The persistent gap between estimated and actual delivery dates arises instead from a structural asymmetry: the costs of underestimating are diffuse and delayed, distributed across a project's eventual overrun, whereas the costs of overestimating are immediate and attributable, since a padded estimate can lose a bid or provoke management scrutiny before the work even begins. Estimators therefore face an incentive gradient that has nothing to do with statistical accuracy and everything to do with organisational consequence, and this gradient reliably pushes estimates toward optimism regardless of how sophisticated the underlying model is. Attempts to correct this by adding explicit contingency buffers frequently fail for a related reason: once a buffer becomes visible in a schedule, it is treated by stakeholders as slack to be reclaimed rather than risk to be reserved, a phenomenon sometimes called the student syndrome, in reference to the tendency to begin a task only as its deadline approaches.
Answer: **(A)** | *Explanation:* Favourable = C(5,2)×C(4,2) = 10×6 = 60. Total ways = C(12,4) = 495. Probability = 60/495 = 4/33....

Q20. A dealer's cost price for a mixture of two rice varieties (mixed 3:2 by weight at ₹40/kg and ₹60/kg) works out to ₹48/kg. He marks up the price by 25% and sells it, but uses a false weight that gives only 900 g for every 1 kg paid for. Find his actual profit percentage.
(a) 38.9%
(b) 35%
(c) 41.7%
(d) 33.3%
Answer: **(A)** | *Explanation:* Marked-up SP for a claimed 1 kg = 48×1.25 = ₹60. True cost of the 900 g actually given = 0.9×48 =
₹43.2. Profit = 60−43.2 = ₹16.8 on a true...

Q21. Pipe A can fill a tank in 12 hours, Pipe B in 15 hours, and a leak can empty the full tank in 20 hours. All three are opened together for 4 hours, after which the leak is plugged. How much more time do A and B together need to fill the remaining tank?
(a) 4 hours
(b) 3 hours
(c) 5 hours
(d) 6 hours
Answer: **(A)** | *Explanation:* Combined rate (A+B−leak) = 1/12+1/15−1/20 = 5/60+4/60−3/60 = 1/10 per hour. In 4 hours, 4/10 =
2/5 of the tank fills, leaving 3/5. A+B's co...

Q22. In a 1000 m race, A beats B by 100 m, and in a separate 1000 m race, B beats C by 100 m (each runner maintains a constant speed across races). By how many metres does A beat C in a 1000 m race between them?
(a) 190 m
(b) 200 m
(c) 180 m
(d) 210 m
Answer: **(A)** | *Explanation:* Speed ratio A:B = 1000:900 = 10:9 (A beats B by 100 m in 1000 m). Similarly B:C = 10:9. So A:C =
(10/9)×(10/9) = 100:81. When A covers 1000...

Q23. An alloy contains 60% copper and 40% zinc. 20 kg of this alloy is melted and mixed with x kg of pure copper to form a new alloy containing 75% copper. Find x.
(a) 12 kg
(b) 10 kg
(c) 15 kg
(d) 8 kg
Answer: **(A)** | *Explanation:* Copper in original alloy = 0.6×20 = 12 kg; zinc = 8 kg (unchanged after adding pure copper). For 75%
copper (25% zinc): 8 = 0.25×(20+x), so...

Q24. ₹8,000 is invested for 1 year at 10% per annum, compounded half-yearly. Find the difference between the compound interest and the simple interest earned on the same sum, at the same rate, for the same period.
(a) ₹20
(b) ₹40

(c) ₹80
(d) ₹100
Answer: **(A)** | *Explanation:* Half-yearly compounding: 2 periods at 5% each. Amount = 8000×1.05×1.05 = 8,820, so CI = ₹820. SI =
8000×10%×1 = ₹800. Difference = 820−800 ...

Q25. In what time will ₹8,000 amount to ₹9,261 at 5% per annum, compounded annually?
(a) 2 years
(b) 3 years
(c) 4 years
(d) 3.5 years

Read the passage and answer Q26–30.
Answer: **(B)** | *Explanation:* 9,261/8,000 = 1.157625 = 1.05^3 (since 1.05^2=1.1025, ×1.05=1.157625). So the time is 3 years.
Section 3: Verbal...

SECTION 3: Verbal Ability (20 min)

Reading Comprehension Passage (Q26–Q30):

```
Software effort estimation has resisted four decades of methodological refinement, and the reason
is not,
as often supposed, a lack of adequate techniques. Function point analysis, story point velocity
tracking
and Monte Carlo simulation are all, in controlled retrospective studies, reasonably well
calibrated. The
persistent gap between estimated and actual delivery dates arises instead from a structural
asymmetry:
the costs of underestimating are diffuse and delayed, distributed across a project's eventual
overrun,
whereas the costs of overestimating are immediate and attributable, since a padded estimate can
lose a
bid or provoke management scrutiny before the work even begins. Estimators therefore face an
incentive
gradient that has nothing to do with statistical accuracy and everything to do with organisational
consequence, and this gradient reliably pushes estimates toward optimism regardless of how
sophisticated the underlying model is. Attempts to correct this by adding explicit contingency
buffers
frequently fail for a related reason: once a buffer becomes visible in a schedule, it is treated
by
stakeholders as slack to be reclaimed rather than risk to be reserved, a phenomenon sometimes
called the
student syndrome, in reference to the tendency to begin a task only as its deadline approaches.
```

Q26. The passage's central claim is that inaccurate software estimates stem primarily from:
(a) Inadequate estimation techniques that fail to model complexity
(b) An organisational incentive structure that rewards optimistic estimates over accurate ones
(c) The inherent unpredictability of software requirements
(d) A lack of contingency buffers in project schedules
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 3, option (B) is directly supported and established by the author's primary thesis ...

Q27. According to the passage, the "student syndrome" refers to:
(a) Estimators lacking sufficient training in statistical methods
(b) Visible schedule buffers being treated as reclaimable slack rather than reserved risk
(c) Junior developers underestimating their own task durations
(d) Students generally being poor at time management
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 3, option (B) is directly supported and established by the author's primary thesis ...

Q28. Which of the following best captures the "structural asymmetry" the author describes?
(a) Overestimation carries immediate, visible costs, while underestimation's costs are delayed and diffuse
(b) Function point analysis is less accurate than story point velocity
(c) Management scrutinises all estimates equally regardless of direction
(d) Contingency buffers eliminate the risk of underestimation
Answer: **(A)** | *Explanation:* According to the reading comprehension passage for Paper 3, option (A) is directly supported and established by the author's primary thesis ...

Q29. The author implies that adding explicit contingency buffers to schedules:
(a) Always improves estimation accuracy
(b) Can be counterproductive once stakeholders perceive the buffer as available slack
(c) Is the only reliable solution to the estimation problem
(d) Has no effect on stakeholder behaviour
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 3, option (B) is directly supported and established by the author's primary thesis ...

Q30. Which statement would the author most likely dispute?
(a) Estimation techniques themselves are the primary obstacle to accurate forecasting
(b) Incentives shape estimation behaviour independent of technique sophistication
(c) A padded estimate can cost a bid before work begins
(d) Underestimation costs tend to surface only over the course of a project
Sentence Correction (Q31–35).
Answer: **(A)** | *Explanation:* According to the reading comprehension passage for Paper 3, option (A) is directly supported and established by the author's primary thesis ...

Q31. Not only was the deployment delayed by unforeseen dependencies, it also required a complete rollback of the previous release.
(a) No improvement
(b) but it also required
(c) also it required
(d) but also it required
Answer: **(B)** | *Explanation:* "Not only...but" needs "but" to complete the correlative pair; "but it also required" is the standard,
grammatical form....

Q32. The findings, which the researchers were confident in it, contradicted the earlier hypothesis.
(a) No improvement
(b) which the researchers were confident about it
(c) in which the researchers were
confident

(d) that the researchers were confident of it
Answer: **(C)** | *Explanation:* The redundant pronoun "it" must be dropped, and the preposition placed before the relative
pronoun: "in which the researchers were confiden...

Q33. Having reviewed the architecture for three days, the decision was made to migrate to microservices.
(a) No improvement
(b) the team decided
(c) it was decided by the team
(d) the migration decision was
made
Answer: **(B)** | *Explanation:* "Having reviewed..." is a dangling modifier — it must describe the people who did the reviewing, so
the sentence needs an explicit human su...

Q34. The new algorithm is more efficient as compared to the previous version.
(a) No improvement
(b) more efficient than
(c) more efficient in comparison of
(d) as more efficient than
Answer: **(B)** | *Explanation:* The correct comparative form is simply "more efficient than", not "as compared to"....

Q35. Having been informed of the outage, immediate action was taken by the on-call engineer.
(a) No improvement
(b) the on-call engineer took immediate action
(c) it was immediately actioned by the
on-call engineer
(d) immediate action, taken by the on-call engineer
Para-jumbles (Q36–38).
Answer: **(B)** | *Explanation:* Another dangling modifier — "Having been informed..." must attach to "the on-call engineer", who
was informed and then acted....

Q36. P: Yet the very flexibility that made the format popular also made it difficult to validate programmatically.
Q: JSON emerged as a lightweight alternative to XML for data interchange between web services.
R: Schema languages such as JSON Schema were developed specifically to address this gap.
S: Unlike XML, it dispensed with the rigid schema definitions many developers found cumbersome.
(a) QSPR
(b) QPSR
(c) SQPR
(d) QSRP
Answer: **(A)** | *Explanation:* Sentence sequence: Q→S→P→R.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q37. P: This is because correlation alone cannot distinguish a cause from a shared underlying factor.
Q: Two variables can move together closely across an entire dataset without either one causing the other.
R: Researchers therefore rely on controlled experiments to isolate genuine causal effects.

S: Ice cream sales and drowning incidents, for instance, both rise in summer, driven by heat rather than by each other.
(a) QSPR
(b) QPSR
(c) SQPR
(d) QPRS
Answer: **(A)** | *Explanation:* Sentence sequence: Q→S→P→R.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q38. P: The resulting backlog of unresolved tickets grew faster than the team could triage it.
Q: The support team initially handled every incoming ticket in the order it arrived.
R: A priority-based queue was introduced only after complaints about critical issues being delayed behind trivial ones.
S: This approach ignored the fact that some issues were far more urgent than others.
(a) QSPR
(b) QPSR
(c) SQPR
(d) PQSR
Vocabulary, Synonyms and Antonyms (Q39–45).
Answer: **(A)** | *Explanation:* Sentence sequence: Q→S→P→R.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q39. The board's decision to proceed despite the warnings was widely seen as obstinate rather than courageous. "Obstinate" means:
(a) flexible
(b) stubbornly unwilling to change
(c) courageous
(d) uncertain
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q40. His perfunctory apology did little to repair the damaged trust. "Perfunctory" means:
(a) heartfelt
(b) done as a mere formality, without real effort
(c) delayed
(d) elaborate
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q41. Fill in the blank: The consultant's report was criticised for being
, offering broad generalisations
instead of concrete recommendations.
(a) meticulous
(b) vague
(c) precise
(d) exhaustive
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q42. Synonym of INSCRUTABLE:
(a) transparent
(b) unfathomable
(c) predictable
(d) simple
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q43. Synonym of PROFLIGATE:
(a) frugal
(b) wasteful
(c) careful
(d) modest
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q44. Antonym of ASSIDUOUS:
(a) diligent
(b) careful
(c) negligent
(d) thorough
Answer: **(C)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (C) is the exact synonym/antonym/definition required in this context....

Q45. Antonym of EPHEMERAL:
(a) fleeting
(b) permanent
(c) brief
(d) transient
Answer: **(B)** | *Explanation:* Section 4: Pseudo Code...

SECTION 4: Pseudo Code (10 min)

Q46. What is printed?
(a) 40
(b) 46
(c) 52
(d) 44

```
int f(int n) {
if (n <= 0) return 0;
if (n == 1) return 1;
return f(n-1) + f(n-2) + n;
}
print(f(6))
```

```
x = 37; mask = 0;
for i = 0 to 5
if (((x >> i) & 1) == 1)
```

```
mask = mask | (1 << (5 - i))
end if
end for
print(mask)
s = 0; i = 1
while (i <= 6)
j = 1
while (j <= i)
if (j % 2 == 0)
j = j + 1
continue
end if
s = s + (i * j)
j = j + 1
end while
i = i + 1
end while
print(s)
push(4); push(7);
a = pop(); b = pop();
push(a - b);
push(10);
c = pop(); d = pop();
push(c + d);
push(3);
e = pop(); f = pop();
print(e * f)
arr = [10, 20, 30, 40, 50, 60]
i = 0; j = 5
while (i < j)
temp = arr[i]
arr[i] = arr[j]
arr[j] = temp
i = i + 1
j = j - 2
end while
print(arr)
```

Answer: **(B)** | *Explanation:* 46. f(0)=0, f(1)=1, f(2)=1+0+2=3, f(3)=3+1+3=7, f(4)=7+3+4=14, f(5)=14+7+5=26, f(6)=26+14+6=46....

Q47. What is printed?

(a) 37

(b) 41

(c) 45

(d) 39

Answer: **(B)** | *Explanation:* 41. 37 = 100101_2 (bits: bit0=1,bit2=1,bit5=1, rest 0). The loop places bit i of x into

position (5−i) of
mask, which reverses the 6-bit pat...

Q48. What is printed?
(a) 130
(b) 124
(c) 136
(d) 118
Answer: **(A)** | *Explanation:* 130. Tracing directly (j increments before each continue, so the loop always terminates correctly):
i=1 contributes 1; i=2 contributes 2; i...

Q49. A stack is used (LIFO). What is printed?
(a) 39
(b) 33
(c) 42
(d) 30
Answer: **(A)** | *Explanation:* 39. Stack: push 4,7 → [4,7]. a=pop=7, b=pop=4 → push(7−4=3) → [3]. push(10) → [3,10]. c=pop=10,
d=pop=3 → push(13) → [13]. push(3) → [13,3]...

Q50. What is the final content of arr? (0-indexed; each statement uses the array's state at that point)
(a) [60, 40, 30, 20, 50, 10]
(b) [60, 20, 30, 40, 50, 10]
(c) [10, 40, 30, 20, 50, 60]
(d) [60, 40, 50, 20, 30, 10]
Answer: **(A)** | *Explanation:* i=0,j=5: swap arr[0],arr[5] → [60,20,30,40,50,10]; i=1,j=3. i<j (1<3): swap arr[1],arr[3] →
[60,40,30,20,50,10]; i=2,j=1. Now i<j fails (2<...

SECTION 5: Numeric Puzzle Solving (10 min)

Q51. Matchstick. A figure is made of 3 unit squares arranged in an L-shape (one square with a second attached to its right, and a third attached below the first), using 10 matchsticks in total. What is the minimum number of sticks to remove so that no complete square remains?
(a) 1
(b) 2
(c) 3
(d) 4
Answer: **(B)** | *Explanation:* 2. The stick shared between the top-left square and the top-right square is also a side of both.
Removing it breaks both squares in one mov...

Q52. Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 5, R = 3, B = 7, L = 2, Centre = 19
Wheel 2: T = 4, R = 6, B = 9, L = 3, Centre = 63
Wheel 3: T = 8, R = 5, B = 6, L = 4, Centre = ?

(a) 54
(b) 58
(c) 62
(d) 50
Answer: **(B)** | *Explanation:* 58. Rule: Centre = (T+B)×(R+L) − (T×B + R×L). Wheel 1: (5+7)(3+2)−(5×7+3×2) = 60−41 = 19 ✓. Wheel
2: (4+9)(6+3)−(4×9+6×3) = 117−54 = 63 ✓. ...

Q53. Matrix logic. Find the missing number.

4
9
25
9
25
49
25
49
?
(a) 100
(b) 121
(c) 144
(d) 81
Answer: **(B)** | *Explanation:* 121. Each entry is the square of a prime, arranged so that reading along each anti-diagonal gives
consecutive primes: $2^2, 3^2, 5^2, 7^2, 11^2$. The ...

Q54. Pattern. Find the next term: 6, 24, 60, 120, 210, ?
(a) 320
(b) 330
(c) 336
(d) 342
Answer: **(C)** | *Explanation:* 336. Each term is the product of 3 consecutive integers: 1×2×3, 2×3×4, 3×4×5, 4×5×6, 5×6×7; next is
6×7×8 = 336.
Section 6: Grammar...

SECTION 6: English Grammar (10 min)

Q55. A number of employees
absent, but the majority of the workforce
present today.
(a) is, is
(b) are, is
(c) is, are
(d) are, are
Answer: **(B)** | *Explanation:* "A number of" + plural noun takes a plural verb (are); "the majority of the

workforce" (workforce
treated as one unit) takes a singular ver...

Q56. By the time she retires next year, she
for this company for over two decades.
(a) will work
(b) will have worked
(c) will have been working
(d) has worked
Answer: **(C)** | *Explanation:* Future perfect continuous ("will have been working") best expresses an ongoing duration ("for over
two decades") continuing up to a future ...

Q57. The proposal was drafted
close consultation
the stakeholders and was later submitted

review.
(a) in, with, for
(b) with, to, in
(c) in, to, for
(d) by, with, to
Answer: **(A)** | *Explanation:* "in close consultation with" ... "submitted for review"....

Q58. Error spoṄng. "The committee not only approved the budget (A) / but also the timeline for implementation (B) / were revised (C) / to reflect the new priorities. (D)"
(a) A
(b) B
(c) C
(d) D
Answer: **(C)** | *Explanation:* The sentence should read "...but also revised the timeline..." to stay parallel with "approved"; as
written, part (C) breaks that parallel ...

Q59. honesty is
best policy, though
history is full of examples where

dishonest have prospered.
(a) An, the, -, the
(b) The, the, -, the
(c) -, the, -, the
(d) An, a, the, an
Answer: **(C)** | *Explanation:* "Honesty" (abstract, general sense) takes no article; "the best policy" (superlative) takes "the";
"history" (general sense) takes no artic...

SECTION 7: English Writing (10 min)

Q60. Write a short technical essay/paragraph of 150–180 words.
Explain, in your own words, the trade-offs between vertical scaling (upgrading a single server's capacity) and horizontal scaling (adding more servers) of a server infrastructure. Recommend which approach a mid-sized e-commerce company experiencing rapid, unpredictable traffic growth should adopt, and justify your recommendation.

Evaluation rubric (10 marks): Technical accuracy of the comparison (3) | Clarity of the recommendation and justification (3) | Structure and coherence (2) | Grammar and vocabulary (1) | Word limit adherence (1)
Answer: **(N/A)** | *Explanation:* Evaluation Rubric (10 Marks): Technical accuracy & depth (3) | Structural coherence (3) | Trade-off synthesis (2) | Grammar & style (1) | Wo...

Mock Exam Paper 04: Infosys System Engineer ITM Mock Paper 4

Source File: `Infosys_SE_Mock_Paper_4.pdf`
Total Questions: 60 Questions | **Total Marks:** 69 Marks (59 MCQs + 10 Marks Writing)

SECTION 1: Reasoning Ability (25 min)

Q01. Statements: Some pens are pencils. All pencils are erasers. No eraser is a sharpener.
Conclusions: I. No pencil is a sharpener. II. Some pens are erasers. III. Some erasers are not pens. IV. Either some pens are sharpeners or no pens are sharpeners.
(a) Only I and II follow
(b) I, II and IV follow
(c) All follow
(d) Only IV follows
Answer: **(B)** | *Explanation:* Pencils⊆erasers and erasers∩sharpeners=∅, so pencils∩sharpeners=∅ (I follows). Some pens are
pencils, and pencils⊆erasers, so some pens are...

Q02. Statements: Only members are allowed entry. Some members are officers. All officers are trained.
Conclusions: I. Some trained persons are members. II. All those allowed entry are trained. III. Some members are trained.
(a) I and III only
(b) II and III only
(c) All follow
(d) None follows
Answer: **(A)** | *Explanation:* "Only members are allowed entry" means allowed-entry⊆members. Some members are officers, and
officers⊆trained, so those members are trained...

Q03. Data Sufficiency. What is the value of x?
I. $x^2 - 5x + 6 = 0$
II. $x > 2$
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not

(c) Both statements together are sufficient, but neither alone is
(d) Even both statements together are not sufficient
Answer: **(C)** | *Explanation:* I alone: $x^2-5x+6=0$ factors to $(x-2)(x-3)=0$, giving $x=2$ or $x=3$ — two solutions, insufficient. II alone: $x>2$
alone is insufficient. Together:...

Q04. Data Sufficiency. In triangle ABC, what is the measure of angle B?
I. Angle A = 50°, and the triangle is isosceles with AB = AC.
II. Angle C = 65°.
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Even both statements together are not sufficient
Answer: **(A)** | *Explanation:* I alone: AB=AC means the angles opposite them are equal, so angle B = angle C; with angle A=50°,
B+C=130°, and B=C gives B=65° — sufficient...

Q05. In a code, each letter is first replaced by its position in the alphabet, and that number is then doubled and reduced by 1. Using this rule, CAT is coded as 5-1-39. What is the code for DOG?
(a) 7-29-13
(b) 8-29-13
(c) 7-30-13
(d) 7-29-14
Answer: **(A)** | *Explanation:* D→4→2(4)−1=7. O→15→2(15)−1=29. G→7→2(7)−1=13. Code: 7-29-13....

Q06. In a code: "alpha beta gamma delta" = "zx qr mn kl", "beta gamma epsilon" = "qr mn vw", "gamma delta zeta" = "mn kl tu", "epsilon zeta eta" = "vw tu yz", "alpha eta theta" = "zx yz op". What is the code for "theta beta zeta"?
(a) op qr tu
(b) op qr vw
(c) zx qr tu
(d) op mn tu
Answer: **(A)** | *Explanation:* Comparing statements pairwise for common words and codes: shines=pu, gamma=mn, bright=mo
(from statements 1&3), giving delta=kl (from state...

Q07. A cube of side 6 units is painted red on all faces and then cut into 1-unit cubes. How many of the small cubes have at least 2 painted faces?
(a) 56
(b) 48
(c) 64
(d) 60
Answer: **(A)** | *Explanation:* For an n×n×n cube, cubes with exactly 2 painted faces (edges) = 12(n−2); with exactly 3 (corners) = 8.
For n=6: edges=12×4=48, corners=8, s...

Q08. How many squares (of all sizes) are there in a 4×4 grid of unit squares?
(a) 30

(b) 25
(c) 36
(d) 20
Answer: **(A)** | *Explanation:* Squares of size k in an n×n grid = $(n-k+1)^2$. For n=4: k=1→16, k=2→9, k=3→4, k=4→1. Total = 16+9+4+1
= 30....

Q09. P is the father of Q. Q is the brother of R. S is the mother of R. T is the husband of S. How is P related to T?
(a) Father
(b) P and T are the same person
(c) Brother-in-law
(d) Cannot be determined
Answer: **(B)** | *Explanation:* Q and R are siblings (Q is R's brother), so S (R's mother) is also Q's mother. P is Q's father, so P and S
are Q's parents — meaning P is S...

Q10. If A + B means A is the father of B, A − B means A is the mother of B, and A × B means A is the son of B, which expression shows that M is the maternal grandfather of N?
(a) M + Q − N
(b) M − Q + N
(c) Q + M − N
(d) M × Q − N
Answer: **(A)** | *Explanation:* M+Q: M is father of Q. Q−N: Q is mother of N. So M is father of Q, who is N's mother — M is N's
maternal grandfather....

Q11. A man walks 14 m East, then 20 m North, then 5 m West, then 8 m South. How far is he from the starting point, and in which direction?
(a) 15 m, North-East
(b) 15 m, North-West
(c) 13 m, North-East
(d) 17 m, North-East
Answer: **(A)** | *Explanation:* Net East-West = 14−5 = 9 m East; net North-South = 20−8 = 12 m North. Distance = $\sqrt{(9^2+12^2)} = \sqrt{225} =$
15 m, direction North-East....

Q12. A ship sails 10 km from port on a bearing of 030° (N30°E), then sails 10 km on a bearing of 150° (S30°E). What is the ship's final position relative to the port?
(a) 10 km due East (bearing 090°)
(b) 10 km due North
(c) 17.3 km, bearing 060°
(d) 10 km, bearing 120°
Answer: **(A)** | *Explanation:* Leg 1 (bearing 030°, 10 km): East = 10 sin30° = 5, North = 10 cos30° = 5√3. Leg 2 (bearing 150°, 10 km):
East = 10 sin150° = 5, North = 10 ...

Q13. Find the next term: 2, 5, 12, 27, 58, ?
(a) 121
(b) 125
(c) 115
(d) 118
Answer: **(A)** | *Explanation:* Differences are 3, 7, 15, 31 — each is 2^n-1 (n=2,3,4,5); next difference is $2^6-1=63$. 58+63 = 121....

Q14. Find the next term: 1, 3, 7, 17, 41, ?
(a) 99
(b) 95
(c) 101
(d) 89
Answer: **(A)** | *Explanation:* Each term follows a(n) = 2×a(n−1) + a(n−2): 2×17+7=41 ✓; next = 2×41+17 = 99....

Q15. Find the next term: Y, V, R, M, ?
(a) G
(b) H
(c) F
(d) I
Answer: **(A)** | *Explanation:* Gaps between letters increase by 1 each time: −3, −4, −5, next −6. M(13)−6 = 7 = G.

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SECTION 2: Technical Ability (Quantitative) (35 min)

Q16. In how many ways can 6 men and 4 women be seated at a round table such that no two women sit together?
(a) 43,200
(b) 86,400
(c) 21,600
(d) 14,400
Answer: **(A)** | *Explanation:* Seat 6 men in a circle: (6−1)! = 120 ways. This creates 6 gaps; choose and arrange 4 women in 4 of
these 6 distinct gaps: P(6,4) = 6×5×4×3 ...

Q17. In how many ways can the letters of the word ORANGE be arranged so that the vowels occupy only the odd positions?
(a) 36
(b) 72
(c) 18
(d) 144
Answer: **(A)** | *Explanation:* ORANGE has 3 vowels (O,A,E) and 3 consonants (R,N,G), matching the 3 odd and 3 even positions
exactly. Vowels in odd positions: 3! = 6. Con...

Q18. Three machines A, B and C produce 25%, 35% and 40% of a factory's output respectively, with defect rates of 5%, 4% and 2%. An item is picked at random and found defective. What is the probability it came from machine C?
(a) 16/69
(b) 8/69
(c) 23/69
(d) 25/69
Answer: **(A)** | *Explanation:* P(D) = 0.25×0.05 + 0.35×0.04 + 0.40×0.02 = 0.0125+0.014+0.008 = 0.0345. P(C|D) = (0.40×0.02)/0.0345 = 0.008/0.0345 = 16/69....

Q19. 5 boys and 3 girls are seated randomly in a row of 8 chairs. What is the probability that all 3 girls sit together?
(a) 3/28
(b) 1/8
(c) 3/56
(d) 5/28
Answer: **(A)** | *Explanation:* Treating the 3 girls as one block: favourable arrangements = 6!×3! = 720×6 = 4,320. Total
arrangements = 8! = 40,320. Probability = 4320/40...

Q20. A trader buys 80 kg of rice at ₹25/kg. He sells 50 kg at a 20% profit and wants his overall profit on all 80 kg to be 15%. At what price per kg must he sell the remaining 30 kg?
(a) ₹26.67
(b) ₹25
(c) ₹28
(d) ₹24
Answer: **(A)** | *Explanation:* Total CP = 80×25 = ₹2,000; target total SP (15% profit) = ₹2,300. SP from the first 50 kg (cost ₹1,250,
at 20% profit) = ₹1,500. Remaining ...

Q21. A, B and C can complete a work alone in 20, 30 and 60 days respectively. All three start together. After 4 days, B leaves; after 3 more days, C also leaves, and A finishes the rest alone. In how many total days is the work completed?
(a) 15 days
(b) 14 days
(c) 16 days
(d) 13 days
Answer: **(A)** | *Explanation:* Rates (LCM 60): A=3/60, B=2/60, C=1/60 per day. First 4 days (all three): 4×6/60=24/60 done. Next 3
days (A+C only): 3×4/60=12/60 done — to...

Q22. Two trains, 300 km apart, move towards each other on the same track. Train A departs at 8:00 AM at 60 km/h; Train B departs at 8:30 AM at 90 km/h. At what time do they meet?
(a) 10:18 AM
(b) 10:12 AM
(c) 10:06 AM
(d) 10:24 AM
Answer: **(A)** | *Explanation:* From 8:00 to 8:30, only A moves: 60×0.5=30 km, leaving 270 km. From 8:30,

combined speed = 150
km/h; time = 270/150 = 1.8 hr = 1 hr 48 min....

Q23. In a two-candidate election, the winner received 60% of votes cast and won by 4,800 votes. If 20% of registered voters did not vote and there were no invalid votes, how many voters were registered?
(a) 30,000
(b) 28,000
(c) 32,000
(d) 24,000
Answer: **(A)** | *Explanation:* Winner's margin (60%−40%=20% of votes cast) = 4,800 → votes cast = 4800/0.20 = 24,000. Since this
is 80% of registered voters: registered =...

Q24. Find the compound interest on ₹12,000 for 1.5 years at 10% per annum, compounded annually (interest for the fractional half-year is simple interest on the amount at the end of the first year).
(a) ₹3,860
(b) ₹3,800
(c) ₹3,960
(d) ₹3,720
Answer: **(A)** | *Explanation:* Year 1: 12000×1.10 = 13,200 (CI so far = 1,200). For the next half-year, SI on 13,200 at 10% for 0.5 yr =
660. Total amount = 13,860; total...

Q25. The compound interest on a sum for 2 years is ₹1,050, and for 3 years is ₹1,655, at the same annual rate. The simple interest on the same sum for 2 years, at the same rate, is ₹1,000. Find the rate of interest per annum.
(a) 10%
(b) 12%
(c) 8%
(d) 15%
Read the passage and answer Q26–30.
Distributed systems theory offers a deceptively simple trilemma: a system cannot simultaneously guarantee consistency, availability and partition tolerance, and must therefore sacrifice one when a network partition occurs. In practice, however, the theorem is more often invoked than correctly applied, because partition tolerance is not actually optional for any system whose nodes communicate over an unreliable network, which is to say nearly all distributed systems built today. The real choice, once partition tolerance is accepted as a structural necessity rather than a design preference, is between consistency and availability during the partition itself, and this choice need not be uniform across an entire system. A payment ledger might reasonably refuse writes during a partition to preserve consistency, accepting temporary unavailability as the lesser cost, while a product catalogue serving read traffic might instead continue answering requests with potentially stale data, judging availability the more valuable property for that workload. Treating the CAP theorem as prescribing a single, system-wide choice, rather than a per-operation or per-service one, is therefore a common and consequential misreading.
Answer: **(A)** | *Explanation:* 3rd-year interest = 1655−1050 = ₹605 — but more directly: $CI_2 - SI_2 = Pr^2$ = 1050−1000 = 50, and SI_2 =
2Pr = 1000 so Pr=500. Dividing, r = 50/...

SECTION 3: Verbal Ability (20 min)

Reading Comprehension Passage (Q26–Q30):

```
Read the passage and answer Q26–30.
Distributed systems theory offers a deceptively simple trilemma: a system cannot simultaneously
guarantee consistency, availability and partition tolerance, and must therefore sacrifice one when
a
network partition occurs. In practice, however, the theorem is more often invoked than correctly
applied,
because partition tolerance is not actually optional for any system whose nodes communicate over
an
unreliable network, which is to say nearly all distributed systems built today. The real choice,
once
partition tolerance is accepted as a structural necessity rather than a design preference, is
between
consistency and availability during the partition itself, and this choice need not be uniform
across an
entire system. A payment ledger might reasonably refuse writes during a partition to preserve
consistency, accepting temporary unavailability as the lesser cost, while a product catalogue
serving read
traffic might instead continue answering requests with potentially stale data, judging
availability the
more valuable property for that workload. Treating the CAP theorem as prescribing a single,
system-wide

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choice, rather than a per-operation or per-service one, is therefore a common and consequential
misreading.
```

Q26. According to the passage, the CAP theorem is frequently misapplied because:
(a) Partition tolerance is wrongly treated as an optional design choice when it is effectively mandatory
(b) Consistency is always more important than availability
(c) The theorem does not apply to payment systems
(d) Network partitions rarely occur in practice
Answer: **(A)** | *Explanation:* According to the reading comprehension passage for Paper 4, option (A) is directly supported and established by the author's primary thesis ...

Q27. The payment-ledger versus product-catalogue example illustrates that:
(a) Payment systems should never use distributed architecture
(b) The consistency-availability trade-off can be made differently for different parts of the same overall system
(c) Product catalogues are inherently less important than payment systems
(d) Availability should always be prioritised over consistency
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 4, option (B) is directly supported and established by the author's primary thesis ...

Q28. The phrase "a common and consequential misreading" refers to:
(a) Confusing consistency with availability
(b) Treating the CAP choice as one uniform, system-wide decision rather than one that can vary per service
(c) Believing that partition tolerance can be sacrificed

(d) Assuming all networks are reliable
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 4, option (B) is directly supported and established by the author's primary thesis ...

Q29. Which of the following would the author most likely agree with?
(a) A single distributed system can legitimately favour consistency for one function and availability for another
(b) The CAP theorem is largely irrelevant to real-world system design
(c) Partition tolerance should be sacrificed whenever possible
(d) All read-heavy services must prioritise consistency
Answer: **(A)** | *Explanation:* According to the reading comprehension passage for Paper 4, option (A) is directly supported and established by the author's primary thesis ...

Q30. Based on the passage, why is partition tolerance called a "structural necessity"?
(a) Because regulations require it
(b) Because any system communicating over an unreliable network cannot realistically avoid the possibility of partitions
(c) Because consistency is always more important
(d) Because availability is always more important
Sentence Correction (Q31–35).
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 4, option (B) is directly supported and established by the author's primary thesis ...

Q31. It is imperative that every deployment is tested thoroughly before being pushed to production.
(a) No improvement
(b) be tested
(c) was tested
(d) will be tested
Answer: **(B)** | *Explanation:* "Imperative that" triggers the subjunctive mood, which uses the base form: "be tested"....

Q32. Rarely a system architect gets the chance to redesign a platform from scratch.
(a) No improvement
(b) does a system architect get
(c) a system architect does get
(d) gets a system
architect
Answer: **(B)** | *Explanation:* "Rarely" at the start of a sentence triggers subject-auxiliary inversion: "does a system architect get"....

Q33. The report's conclusion, that many have questioned it, was based on incomplete data.
(a) No improvement
(b) which many have questioned
(c) that many have questioned
(d) many have
questioned which
Answer: **(B)** | *Explanation:* The redundant "it" must be dropped, and "which" (not "that") is used after a comma

for a non-
restrictive clause....

Q34. The budget overrun was attributed to the fact of poor estimation practices rather than external factors.
(a) No improvement
(b) to poor estimation practices
(c) to the fact that poor estimation practices
(d) because of poor estimation practices
Answer: **(B)** | *Explanation:* "The fact of" is redundant and unidiomatic; "attributed to poor estimation practices" is correct and
concise....

Q35. Being that the deadline was moved up, the team had to prioritise ruthlessly.
(a) No improvement
(b) Given that the deadline was moved up
(c) Being the deadline was moved up
(d) Because of the deadline was moved up
Para-jumbles (Q36–38).
Answer: **(B)** | *Explanation:* "Being that" is non-standard; the correct formal causal connector is "Given that"....

Q36. P: Yet this very redundancy, when poorly managed, can itself become a source of inconsistency.
Q: Distributed databases often replicate data across multiple nodes to improve fault tolerance.
R: Conflict-resolution strategies, such as vector clocks, were developed to reconcile such divergent copies.
S: If two replicas are updated independently before synchronising, they may end up holding conflicting values.
(a) QPSR
(b) QSPR
(c) PQSR
(d) QRPS
Answer: **(A)** | *Explanation:* Sentence sequence: Q→P→S→R.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q37. P: It instead reflects a deliberate trade-off between latency and durability that every system designer must
make explicitly.
Q: Writing data to disk before acknowledging a request guarantees durability but adds latency.
R: Skipping this step and acknowledging immediately improves responsiveness at the risk of data loss during a crash.
S: Neither approach is objectively superior, and the choice is not a matter of engineering carelessness.
(a) QRSP
(b) QSRP
(c) RQSP
(d) QRPS
Answer: **(A)** | *Explanation:* Sentence sequence: Q→R→S→P.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q38. P: This made debugging nearly impossible, since no single node had a complete picture of the request's path.
Q: Early microservice deployments often lacked any form of distributed tracing.
R: Modern observability platforms now stitch together logs from every service a request touches.
S: A single user request might traverse a dozen services before returning a response.
(a) QSPR
(b) QPSR
(c) SQPR
(d) QSRP
Vocabulary, Synonyms and Antonyms (Q39–45).
Answer: **(A)** | *Explanation:* Sentence sequence: Q→S→P→R.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q39. The auditor's findings were unequivocal, leaving no room for alternative interpretation. "Unequivocal" means:
(a) ambiguous
(b) leaving no doubt; completely clear
(c) tentative
(d) biased
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q40. Critics dismissed the proposal as specious — appealing on the surface but fundamentally flawed. "Specious" means:
(a) genuinely sound
(b) superficially plausible but actually wrong
(c) deeply researched
(d) widely accepted
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q41. Fill in the blank: Despite years of effort, the two departments remained _____ in their approaches, unable to find common ground.
(a) aligned
(b) intransigent
(c) harmonious
(d) cooperative
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q42. Synonym of CIRCUMSPECT:
(a) reckless
(b) cautious and careful
(c) indifferent
(d) impulsive
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q43. Synonym of OBFUSCATORY:
(a) clarifying
(b) deliberately confusing
(c) simplifying
(d) transparent
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q44. Antonym of PARSIMONIOUS:
(a) frugal
(b) stingy
(c) lavish
(d) economical
Answer: **(C)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (C) is the exact synonym/antonym/definition required in this context....

Q45. Antonym of GARRULOUS:
(a) talkative
(b) chatty
(c) taciturn
(d) verbose
Answer: **(C)** | *Explanation:* Infosys System Engineer |Mock Test
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Section 4: Pseudo Code...

SECTION 4: Pseudo Code (10 min)

Q46. What is printed? (% is the modulo operator)
```
int gcd(int a, int b) {
if (b == 0) return a;
return gcd(b, a % b);
}
print(gcd(252, 105))
```
(a) 21
(b) 42
(c) 7
(d) 14
Answer: **(A)** | *Explanation:* 21. gcd(252,105): 252%105=42 → gcd(105,42): 105%42=21 → gcd(42,21): 42%21=0 → gcd(21,0)=21....

Q47. What is printed? (^=XOR, |=OR, &=AND, >>=right shift)
```
a = 13; b = 25;
c = (a ^ b) & (a | b);
d = c >> 2;
print(c + d)
```
(a) 25
(b) 29
(c) 20

(d) 33

Answer: **(A)** | *Explanation:* 25. 13=01101, 25=11001. a^b = 10100 = 20. a|b = 11101 = 29. c = 20 & 29 = 10100 = 20. d = 20>>2 =
5. c+d = 20+5 = 25....

Q48. How many swaps are performed in total?

```
arr = [5, 2, 8, 1, 9, 3]
swaps = 0
for i = 0 to 4
for j = 0 to 4 - i
if (arr[j] > arr[j+1])
temp = arr[j]
arr[j] = arr[j+1]
arr[j+1] = temp
swaps = swaps + 1
end if
end for
end for
print(swaps)
```

(a) 7

(b) 8

(c) 6

(d) 9

Answer: **(A)** | *Explanation:* 7. Full bubble-sort trace: pass i=0 gives 3 swaps (array becomes [2,5,1,8,3,9]); pass i=1 gives 2 more
swaps ([2,1,5,3,8,9]); pass i=2 give...

Q49. This algorithm reverses the first k elements of a queue using a stack. What is printed?

```
queue = [1, 2, 3, 4, 5]
stack = []
k = 3
for i = 1 to k
push(stack, dequeue(queue))
end for
while (stack is not empty)
enqueue(queue, pop(stack))
end while
for i = 1 to (5 - k)
enqueue(queue, dequeue(queue))
end for
print(queue)
```

(a) [3, 2, 1, 4, 5]

(b) [1, 2, 3, 4, 5]

(c) [5, 4, 3, 2, 1]

(d) [3, 2, 1, 5, 4]

Answer: **(A)** | *Explanation:* This is the standard "reverse first k of a queue" algorithm. After moving the first 3

elements through
the stack, the queue becomes [4,5,3,...

Q50. This algorithm moves all non-zero elements to the front, preserving order, and fills the rest with zeroes. What is the final arr?
arr = [3, 0, 4, 0, 5, 0, 2]
pos = 0
for i = 0 to 6
if (arr[i] != 0)
arr[pos] = arr[i]
pos = pos + 1
end if
end for
while (pos <= 6)
arr[pos] = 0
pos = pos + 1
end while
print(arr)
(a) [3, 4, 5, 2, 0, 0, 0]
(b) [3, 0, 4, 0, 5, 0, 2]
(c) [3, 4, 5, 2, 0, 0, 2]
(d) [0, 0, 0, 3, 4, 5, 2]
Answer: **(A)** | *Explanation:* Non-zero elements (3,4,5,2) are copied to the front in order, giving positions 0–3; the while loop then
zeroes out positions 4,5,6. Final: ...

SECTION 5: Numeric Puzzle Solving (10 min)

Q51. Matchstick. A 3×3 grid of unit squares (9 small squares) is built using 24 matchsticks. Counting squares of every size (1×1, 2×2 and 3×3 — 14 squares in total), what is the minimum number of matchsticks that must be removed so that no complete square of any size remains?
(a) 4
(b) 5
(c) 6
(d) 7
Answer: **(C)** | *Explanation:* 6. This was verified by exhaustive computer search over all 24 sticks and all 14 squares (nine 1×1, four
2×2, one 3×3): no set of 5 or fewe...

Q52. Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 4, R = 5, B = 6, L = 3, Centre = 39
Wheel 2: T = 7, R = 2, B = 8, L = 5, Centre = 66
Wheel 3: T = 6, R = 4, B = 9, L = 2, Centre = ?
(a) 58
(b) 62
(c) 66

(d) 54

Answer: **(B)** | *Explanation:* 62. Rule: Centre = (T+R)×(B+L) − (T×L + R×B). Wheel 1: (4+5)(6+3)−(4×3+5×6) = 81−42 = 39 ✓. Wheel 2: (7+2)(8+5)−(7×5+2×8) = 117−51 = 66 ✓. ...

Q53. Matrix logic. Find the missing number.

2

3

4

6

8

10

12

15

?

(a) 16

(b) 18

(c) 20

(d) 21

Answer: **(B)** | *Explanation:* 18. Rule: cell(row r, col c) = r×c + r^2 (rows and columns numbered from 1). Row 3: (3×1+9, 3×2+9, 3×3+9) = (12, 15, 18)....

Q54. Pattern. Find the next term: 1, 5, 14, 30, 55, ?

(a) 91

(b) 85

(c) 96

(d) 90

Answer: **(A)** | *Explanation:* 91. Each term is the sum of squares $1^2+2^2+...+n^2$: 1, 1+4=5, +9=14, +16=30, +25=55, next +36 = 91.

Section 6: Grammar...

SECTION 6: English Grammar (10 min)

Q55. _____ the system crash occurred during peak hours, the financial losses would have been catastrophic.

(a) If

(b) Had

(c) Should

(d) Unless

Answer: **(B)** | *Explanation:* Inverted third conditional (omitting "if"): "Had the crash occurred..." = "If the crash had occurred..."....

Q56. The engineer explained that the server _____ down due to a memory leak that _____ undetected for weeks.

(a) went, had gone

(b) had gone, went

(c) goes, had gone

(d) went, has gone
Answer: **(A)** | *Explanation:* The main reported event ("went down") takes simple past; the leak's undetected state, which existed
before and up to that event, takes past...

Q57. The team is responsible _____ delivering the project _____ budget and _____ the agreed deadline.
(a) for, within, by
(b) of, under, before
(c) for, under, within
(d) to, in, at
Answer: **(A)** | *Explanation:* "responsible for" ... "within budget" ... "by the deadline"....

Q58. Error spotting. "The new system is not only faster than the old one (A) / but also requires (B) / less maintenance, making it (C) / a more cost-effective to operate. (D)"
(a) A
(b) B
(c) C
(d) D
Answer: **(D)** | *Explanation:* "cost-effective" is an adjective, not a noun, so "a more cost-effective to operate" is ungrammatical —
the error is in part (D)....

Q59. _____ information technology has transformed _____ way we work, though _____ access to _____ internet remains unequal across regions.
(a) -, the, -, the
(b) The, the, an, the
(c) -, the, the, the
(d) An, the, -, an
Answer: **(A)** | *Explanation:* "Information technology" (general/abstract) takes no article; "the way we work" takes "the" (specific,
fixed phrase); "access" (general) ta...

SECTION 7: English Writing (10 min)

Q60. Write a short technical essay/paragraph of 150–180 words.
A legacy monolithic application your team supports is becoming difficult to maintain and scale. Explain the key risks of migrating it to a microservices architecture versus keeping it monolithic but modularising the codebase internally, and recommend one approach for a team of 8 engineers with limited DevOps experience, justifying your choice.
Evaluation rubric (10 marks): Technical accuracy of the comparison (3) | Clarity and soundness of the recommendation given the team's constraints (3) | Structure and coherence (2) | Grammar and vocabulary (1) | Word limit adherence (1)
Answer: **(N/A)** | *Explanation:* Evaluation Rubric (10 Marks): Technical accuracy (3) | Risk analysis (3) | Phasing justification (2) | Vocabulary & grammar (1) | Word limit...

Mock Exam Paper 05: Infosys System Engineer ITM Mock Paper 5

Source File: `Infosys_SE_Mock_Paper_5.pdf`
Total Questions: 60 Questions | **Total Marks:** 69 Marks (59 MCQs + 10 Marks Writing)

SECTION 1: Reasoning Ability (25 min)

Q01. Statements: All toys are games. All games are activities. Some crafts are not activities. No craft is a hobby.
Conclusions: I. All toys are activities. II. Some crafts are not hobbies. III. No toy is a craft.
(a) Only I follows
(b) I and II follow
(c) I, II and III follow
(d) None follows
Answer: **(B)** | *Explanation:* Toys⊆games⊆activities, so all toys are activities (I follows). "Some crafts are not activities" confirms
crafts exist, and since no craft i...

Q02. Statements: All researchers are scholars. No scholar is lazy. Some lazy people are students. All students are young.
Conclusions: I. No researcher is lazy. II. Some students are not scholars. III. Some young people are lazy.
(a) Only I and II follow
(b) Only II and III follow
(c) All follow
(d) None follows
Answer: **(C)** | *Explanation:* Researchers⊆scholars, and scholars∩lazy=∅, so researchers∩lazy=∅ (I). Some lazy are students, and
since scholars∩lazy=∅, those lazy-student...

Q03. Data Sufficiency. If x and y are positive integers, what is the remainder when x is divided by y?
I. x = 5y + 3
II. y = 7
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Even both statements together are not sufficient
Answer: **(C)** | *Explanation:* I alone: x=5y+3 only gives remainder 3 if y>3; for small y (e.g., y=2, x=13, 13÷2 leaves remainder 1, not
3) it fails — insufficient alone....

Q04. Data Sufficiency. In how many days can A and B complete a job together?
I. A is twice as efficient as B.
II. A alone can complete the job in 12 days.
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Even both statements together are not sufficient
Answer: **(C)** | *Explanation:* II alone gives only A's rate (1/12), with B's rate unknown. I alone gives only a ratio

(2:1), with no
absolute rate. Together: A's rate=1/1...

Q05. In a code, letters in odd positions of a word are shifted 2 places forward in the alphabet, and letters in even positions are shifted 1 place forward. Using this rule, SYSTEM is coded as UZUUGN. What is the code for SERVER?
(a) UFTWGS
(b) UFTWFS
(c) UGTWGS
(d) UFUWGS
Answer: **(A)** | *Explanation:* S(odd,+2)=U. E(even,+1)=F. R(odd,+2)=T. V(even,+1)=W. E(odd,+2)=G. R(even,+1)=S. Code: UFTWGS....

Q06. In a code: "data packet loss" = "zt rp qm", "loss causes retransmission" = "qm xy lp", "retransmission increases latency" = "lp wv ou", "latency affects data" = "ou nk zt", "packet causes latency" = "rp xy ou". What is the code for "increases affects packet"?
(a) wv nk rp
(b) wv nk qm
(c) ou nk rp
(d) wv xy rp
Answer: **(A)** | *Explanation:* Pairwise comparison of the 5 statements resolves every word: data=zt, packet=rp, loss=qm, causes=xy,
retransmission=lp, increases=wv, laten...

Q07. A rectangular cuboid is cut by 2 parallel cuts along its length, 1 cut along its breadth, and 3 cuts along its height (each cut perpendicular to that dimension), producing smaller cuboids from an originally fully-painted surface. How many of the resulting small cuboids have exactly 1 painted face?
(a) 4
(b) 6
(c) 8
(d) 2
Answer: **(A)** | *Explanation:* With 2 cuts in length (3 pieces), 1 cut in breadth (2 pieces), 3 cuts in height (4 pieces), every piece
touches an outer breadth-face (sinc...

Q08. How many rectangles (of all sizes, including squares) are there in a 4×4 grid of unit squares?
(a) 100
(b) 90
(c) 81
(d) 110
Answer: **(A)** | *Explanation:* Choosing 2 of the 5 horizontal lines and 2 of the 5 vertical lines of a 4×4 grid gives a rectangle:
C(5,2)×C(5,2) = 10×10 = 100....

Q09. P and Q are a married couple. R is the son of P and Q. S is the daughter of P and Q. R is married to T. U is
the son of R and T. V is the daughter of S. How is U related to V?
(a) Cousins

(b) Siblings
(c) Nephew and Niece
(d) Uncle and Niece
Answer: **(A)** | *Explanation:* R and S are siblings (children of P and Q). U is R's son; V is S's daughter. Since their parents (R and S)
are siblings, U and V are cousin...

Q10. If A + B means A is the father of B, A − B means A is the mother of B, A × B means A is the son of B, and A ÷ B means A is the wife of B, which expression shows that M is the father-in-law of N (N being male)?
(a) M + Q ÷ N
(b) M ÷ Q + N
(c) Q + M ÷ N
(d) M × Q ÷ N
Answer: **(A)** | *Explanation:* M+Q: M is father of Q. Q÷N: Q is wife of N. So M is father of N's wife — M is N's father-in-law....

Q11. A man starts from point O and walks 10 m North-East, then 10 m North-West, then 10 m South-West. How far is he from O, and in which direction?
(a) 10 m, North-West
(b) 10 m, South-West
(c) 14.14 m, North-West
(d) 5 m, North-West
Answer: **(A)** | *Explanation:* Decomposing each 10 m leg into East/North components (each at 45°, so ±5√2 in each axis): totals
work out to East=−5√2, North=+5√2, giving ...

Q12. A man is initially facing a bearing of 040°. After making exactly 2 equal clockwise turns, he ends up facing a bearing of 310°. What was the magnitude of each turn?
(a) 135°
(b) 142.5°
(c) 120°
(d) 150°
Answer: **(A)** | *Explanation:* Total clockwise rotation needed = 310°−40° = 270°. Split equally over 2 turns: 270/2 = 135° each
(verified: 40°+135°=175°, 175°+135°=310° ✓...

Q13. Find the next term: 5, 14, 41, 122, 365, ?
(a) 1094
(b) 1095
(c) 1093
(d) 1096
Answer: **(A)** | *Explanation:* Each term follows a(n) = 3×a(n−1) − 1: 122×3−1=365 ✓; next = 365×3−1 = 1094....

Q14. Find the next term in this 7-term series: 4, 9, 7, 16, 10, 25, 13, ?
(a) 36
(b) 16
(c) 49
(d) 30

Answer: **(A)** | *Explanation:* This series interleaves two sequences: odd positions (4,7,10,13,...) increase by 3; even positions
(9,16,25,...) are 3^2,4^2,5^2. The 8th term...

Q15. Find the next term: D, H, M, S, Z, ?
(a) H
(b) G
(c) I
(d) F
Answer: **(A)** | *Explanation:* Gaps increase by 1 each time: +4,+5,+6,+7; next gap +8. Z(26)+8=34, which wraps around (34−26=8)
to H.

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SECTION 2: Technical Ability (Quantitative) (35 min)

Q16. A committee of 5 is to be formed from 6 men and 5 women. A specific man X must be included, a specific woman Y must be excluded, and the committee must include at least 1 woman. In how many ways can this be done?
(a) 121
(b) 126
(c) 120
(d) 115
Answer: **(A)** | *Explanation:* With X fixed in and Y fixed out, the remaining pool is 5 men + 4 women = 9 people, from which 4
more members are chosen: C(9,4)=126. Subtra...

Q17. In how many ways can the 8 distinct letters of COMPUTER be arranged such that C and T have exactly 2 letters between them (in either order)?
(a) 7200
(b) 6000
(c) 8400
(d) 5040
Answer: **(A)** | *Explanation:* Valid position-pairs with C and T exactly 3 apart: 5 pairs with C before T, plus 5 with T before C = 10
total. The remaining 6 distinct let...

Q18. Two tickets are drawn one after another, without replacement, from a box containing tickets numbered 1 to 10. What is the probability that the number on the second ticket is greater than the number on the first?
(a) 1/2
(b) 9/20
(c) 11/20
(d) 2/5
Answer: **(A)** | *Explanation:* By symmetry, for any two distinct numbers drawn in some order, exactly half the time the second
exceeds the first. Probability = 1/2 (confi...

Q19. A mini-deck has 16 cards: 4 suits with 4 cards each. 4 cards are drawn at random. What is the probability that at least 2 of them share the same suit?
(a) 391/455
(b) 64/455
(c) 327/455
(d) 201/455
Answer: **(A)** | *Explanation:* P(all 4 different suits) = (4×4×4×4)/C(16,4) = 256/1820 = 64/455. P(at least 2 same suit) = 1−64/455 =
391/455....

Q20. A shopkeeper runs a "Buy 3, Get 1 Free" offer on an item marked at ₹200 (costing him ₹120 each). A customer buys 4 items under this offer, paying for 3. What is the shopkeeper's profit percentage, based on his cost for all 4 items given?
(a) 25%
(b) 20%
(c) 30%
(d) 15%
Answer: **(A)** | *Explanation:* Cost for 4 items = 4×120 = ₹480. Revenue (customer pays for 3 at ₹200) = ₹600. Profit = ₹120 on a
cost of ₹480, giving 120/480×100 = 25%....

Q21. A, B and C can individually complete a work in 15, 20 and 30 days. They work together, and the total payment of ₹9,000 is distributed in proportion to each person's work contribution. How much should C be paid?
(a) ₹2,000
(b) ₹1,800
(c) ₹2,200
(d) ₹2,500
Answer: **(A)** | *Explanation:* Rates in the ratio 1/15:1/20:1/30 = 4:3:2 (over LCM 60). C's share = (2/9)×9000 = ₹2,000....

Q22. Two runners start from the same point on a 600 m circular track at the same time, in opposite directions, at 8 m/s and 10 m/s. After how many seconds will they both simultaneously be back at the starting point for the first time?
(a) 300 s
(b) 200 s
(c) 150 s
(d) 600 s
Answer: **(A)** | *Explanation:* A takes 600/8=75 s/lap; B takes 600/10=60 s/lap. Both are back at the start simultaneously at
LCM(75,60) = 300 s....

Q23. A container has 80 litres of pure milk. 20 litres of the mixture is removed and replaced with water; this process is repeated once more. How much pure milk remains?
(a) 45 litres
(b) 50 litres
(c) 40 litres
(d) 48 litres

Answer: **(A)** | *Explanation:* Using the repeated-dilution formula: Final = $80\times(1-20/80)^2 = 80\times(3/4)^2 = 80\times9/16$ = 45 litres....

Q24. At a certain rate of annual compound interest, a sum doubles itself in 5 years. In how many years will it become 8 times itself, at the same rate?
(a) 15 years
(b) 20 years
(c) 10 years
(d) 25 years
Answer: **(A)** | *Explanation:* $(1+r)^5=2$ (doubling). For 8×: $(1+r)^n=8=2^3=[(1+r)^5]^3=(1+r)^{15}$, so n=15 years....

Q25. ₹20,000 is borrowed at 10% per annum compound interest (annual compounding). ₹11,000 is repaid at the end of year 1. What amount must be paid at the end of year 2 to clear the entire debt?
(a) ₹12,100
(b) ₹12,000
(c) ₹11,500
(d) ₹13,200
Read the passage and answer Q26–30.
Rate limiting is often presented as a purely defensive mechanism, a way to shield backend services from abusive or accidental traffic spikes, but this framing understates its role as a tool for communicating capacity contracts between services. A token-bucket limiter that rejects requests once a quota is exhausted is, in effect, declaring an explicit promise about how much load a downstream service can absorb, and client-side code written against that promise can make informed decisions about retry timing, backoff, and degraded functionality rather than failing unpredictably. The trouble arises when limits are set reactively, in response to an incident, rather than derived from actual capacity testing; a

limit chosen under pressure to stop an ongoing outage often reflects the conditions of that specific incident rather than the service's true sustainable throughput, and may be far too conservative once normal operation resumes, quietly capping legitimate growth long after the original emergency has been forgotten. Treating rate limits as living documentation of measured capacity, revisited whenever the underlying service's performance characteristics change, avoids both failure modes: the limiter protects the service without becoming an invisible ceiling on its own success.
Answer: **(A)** | *Explanation:* End of year 1: 20000×1.10=22,000; after repaying 11,000, remaining debt = 11,000. End of year 2:
11000×1.10 = ₹12,100.
Section 3: Verbal...

SECTION 3: Verbal Ability (20 min)

Reading Comprehension Passage (Q26–Q30):

```
Read the passage and answer Q26–30.
Rate limiting is often presented as a purely defensive mechanism, a way to shield backend services
from
abusive or accidental traffic spikes, but this framing understates its role as a tool for
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capacity contracts between services. A token-bucket limiter that rejects requests once a quota is
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specific
incident rather than the service's true sustainable throughput, and may be far too conservative
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normal operation resumes, quietly capping legitimate growth long after the original emergency has
been
forgotten. Treating rate limits as living documentation of measured capacity, revisited whenever
the
underlying service's performance characteristics change, avoids both failure modes: the limiter
protects
the service without becoming an invisible ceiling on its own success.
```

Q26. The passage's central argument is that rate limits should be understood primarily as:
(a) A purely defensive mechanism against malicious traffic
(b) An explicit, measured capacity contract that client code can plan around, not just a defensive wall
(c) A permanent configuration that should never be revisited
(d) A substitute for proper capacity testing
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 5, option (B) is directly supported and established by the author's primary thesis ...

Q27. According to the passage, limits set reactively during an incident risk becoming:
(a) Too generous, allowing further outages
(b) An invisible ceiling that quietly caps legitimate growth after the emergency has passed
(c) Immediately replaced by automated systems
(d) Irrelevant to client-side retry logic
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 5, option (B) is directly supported and established by the author's primary thesis ...

Q28. The phrase "living documentation of measured capacity" implies that rate limits should be:
(a) Set once and never changed
(b) Periodically revisited as the service's performance characteristics evolve
(c) Removed once a service proves reliable
(d) Determined solely by client demand
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 5, option (B) is directly supported and established by the author's primary thesis ...

Q29. Which of the following would the author most likely agree with?
(a) A well-chosen rate limit communicates useful information to the systems that call it, beyond just blocking excess traffic
(b) Rate limits should always be set as conservatively as possible
(c) Capacity testing is unnecessary if a rate limiter is in place
(d) Client-side retry logic is unrelated to how a rate limit is set
Answer: **(A)** | *Explanation:* According to the reading comprehension passage for Paper 5, option (A) is directly supported and established by the author's primary thesis ...

Q30. Based on the passage, what is the main risk of NOT revisiting a reactively-set limit?
(a) The service becomes vulnerable to outages again
(b) The limit may stay more conservative than necessary and suppress legitimate growth
(c) Client code will stop retrying altogether
(d) The token-bucket algorithm will stop functioning
Sentence Correction (Q31–35).
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 5, option (B) is directly supported and established by the author's primary thesis ...

Q31. If the migration was properly tested, the system would not be experiencing intermittent failures right now.
(a) No improvement
(b) had been, would not be
(c) were, will not be
(d) has been, would not have been
Answer: **(B)** | *Explanation:* A mixed conditional (past condition, present result) needs "had been" (past perfect) in the if-clause
and "would not be" (present condition...

Q32. The lead engineer said that the deployment will take place the following week, as the testing has not yet been completed at that time.
(a) No improvement
(b) would, had
(c) would, has
(d) will, had
Answer: **(A)** | *Explanation:* Reported speech backshifts "will"→"would" and "has not been completed"→"had not been
completed"....

Q33. The team had to account for the discrepancy, which they eventually traced to a configuration error introduced by mistake during the last deployment.
(a) No improvement
(b) of, to, in
(c) with, into, by
(d) for, into, with
Answer: **(A)** | *Explanation:* "account for", "traced to", and "by mistake" are the correct idiomatic combinations....

Q34. The new policy requires all employees to attending mandatory security training within thirty days of joining, failing which their system access will be automatically suspended.

(a) No improvement
(b) attending
(c) to attend
(d) attend
Answer: **(C)** | *Explanation:* "requires...to attend" needs the infinitive, not the gerund ("to attending" is ungrammatical)....

Q35. The outage was caused due to a misconfigured load balancer that had been deployed without peer review.
(a) No improvement
(b) was caused by
(c) was caused because of
(d) caused due to
Para-jumbles (Q36–38).
Answer: **(B)** | *Explanation:* "caused by" is correct; "caused due to" is a common but non-standard (redundant) construction....

Q36. P: This made it difficult to reason about which version of a shared object any given thread would observe.
Q: Early concurrent programming models allowed any thread to read or write shared memory at any time.
R: Memory models that formally define visibility and ordering guarantees were developed to resolve this ambiguity.
S: Without explicit synchronisation, two threads could see updates to the same variable in different orders.
(a) QSPR
(b) QPSR
(c) SQPR
(d) QSRP
Answer: **(A)** | *Explanation:* Sentence sequence: Q→S→P→R.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q37. P: A cache invalidated too eagerly provides little performance benefit, defeating its own purpose.
Q: Cache invalidation is frequently cited as one of the two genuinely hard problems in computer science.
R: One invalidated too late serves stale data that may silently corrupt downstream computations.
S: The difficulty lies in striking a balance between staleness and overhead that rarely has a universally correct answer.
(a) QPRS
(b) QRPS
(c) QPSR
(d) PQRS
Answer: **(A)** | *Explanation:* Sentence sequence: Q→P→R→S.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q38. P: Engineers began instrumenting every service boundary to capture latency at each hop.
Q: A single slow request in a microservice architecture can be caused by any one of a dozen interacting services.
R: Distributed tracing tools emerged to stitch these individual measurements into one coherent timeline.
S: Without this visibility, a single slow endpoint could remain unexplained for weeks.
(a) QSPR

(b) QPSR
(c) SQPR
(d) QSRP
Vocabulary, Synonyms and Antonyms (Q39–45).
Answer: **(A)** | *Explanation:* Sentence sequence: Q→S→P→R.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q39. The consultant's recommendations were anodyne, offering nothing that would upset any stakeholder. "Anodyne" means:
(a) controversial
(b) inoffensive and bland
(c) insightful
(d) aggressive
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q40. The engineer's terse response left no room for follow-up questions. "Terse" means:
(a) elaborate
(b) brief and to the point, almost curt
(c) friendly
(d) confusing
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q41. Fill in the blank: The postmortem was praised for being _____ rather than accusatory, focusing on systemic causes instead of individual blame.
(a) punitive
(b) blameless
(c) hostile
(d) dismissive
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q42. Synonym of INTRACTABLE:
(a) manageable
(b) difficult to control or solve
(c) flexible
(d) simple
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q43. Synonym of PRECARIOUS:
(a) stable
(b) unstable or risky
(c) secure
(d) permanent
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q44. Antonym of MERITORIOUS:
(a) praiseworthy
(b) commendable
(c) blameworthy
(d) deserving
Answer: **(C)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (C) is the exact synonym/antonym/definition required in this context....

Q45. Antonym of VINDICATE:
(a) justify
(b) exonerate
(c) incriminate
(d) clear
Answer: **(C)** | *Explanation:* Section 4: Pseudo Code...

SECTION 4: Pseudo Code (10 min)

Q46. What is printed? (This computes the number of grid paths from one corner to another, moving only right or down.)

```
int f(int n, int m) {
if (n == 0 || m == 0) return 1;
return f(n-1, m) + f(n, m-1);
}
print(f(3, 2))
```

(a) 10
(b) 8
(c) 12
(d) 9
Answer: **(A)** | *Explanation:* 10. This recursion counts grid paths and equals C(n+m,n). Tracing directly: f(1,1)=2, f(2,1)=3, f(1,2)=3,
f(2,2)=6, f(3,1)=4, f(3,2)=f(2,2)...

Q47. What is printed? (>>=right shift, <<=left shift, &=AND, ^=XOR)

```
x = 0;
for i = 0 to 7
if (((45 >> i) & 1) == 1)
x = x + (1 << (7 - i))
end if
end for
y = x ^ 45
print(y)
```

(a) 153
(b) 180
(c) 135
(d) 171
Answer: **(A)** | *Explanation:* 153. 45 = 00101101 in 8 bits. The loop reverses these 8 bits into x, giving x=180

(10110100). y = x^45
= 180^45 = 153 (verified by direct c...

Q48. This is a binary search for target = 50 in a sorted array. How many comparisons are made before the loop ends (the target is not present)?
arr = [2, 5, 8, 12, 16, 23, 38, 45, 56, 72, 91]
low = 0; high = 10; target = 50; comparisons = 0
while (low <= high)
mid = (low + high) / 2
comparisons = comparisons + 1
if (arr[mid] == target)
break
else if (arr[mid] < target)
low = mid + 1
else
high = mid - 1
end if
end while
print(comparisons)
(a) 4
(b) 3
(c) 5
(d) 6
Answer: **(A)** | *Explanation:* 4. Tracing the search for target=50 (not present): mid=5(23)→low=6;
mid=8(56)→high=7;
mid=6(38)→low=7; mid=7(45)→low=8, loop ends. That's 4...

Q49. This stack trace evaluates a postfix-style expression. What is printed?
push(6); push(2); push(3);
c = pop(); b = pop(); push(b + c);
d = pop(); e = pop(); push(e - d);
push(3); push(8); push(2);
f = pop(); g = pop(); push(g / f);
h = pop(); i = pop(); push(i + h);
j = pop(); k = pop(); push(k * j);
print(pop())
(a) 7
(b) 10
(c) 14
(d) 4
Answer: **(A)** | *Explanation:* 7. This performs the postfix evaluation of "6 2 3 + − 3 8 2 / + ×": (2+3)=5, (6−5)=1, (8/2)=4, (3+4)=7,
(1×7)=7....

Q50. This is Kadane's algorithm for maximum subarray sum. What is printed?
arr = [-2, 3, -1, 5, -3, 4, -2, 6]
maxSoFar = arr[0]; maxEndingHere = arr[0]

for i = 1 to 7
maxEndingHere = max(maxEndingHere + arr[i], arr[i])
maxSoFar = max(maxSoFar, maxEndingHere)
end for
print(maxSoFar)
(a) 12
(b) 10
(c) 9
(d) 15
Answer: **(A)** | *Explanation:* 12. Kadane's algorithm finds the maximum subarray sum. Tracing: maxEndingHere/maxSoFar
progress to 3, 2→3, 7, 4→7, 8, 6→8, 12→12 (the subar...

SECTION 5: Numeric Puzzle Solving (10 min)

Q51. Matchstick. Five unit squares are arranged in a plus/cross pattern (one centre square with one square attached to each of its four sides), using 16 matchsticks in total. What is the minimum number of sticks that must be removed so that no complete square remains?
(a) 3
(b) 4
(c) 5
(d) 6
Answer: **(B)** | *Explanation:* 4. Each of the 4 outer squares shares exactly one stick with the centre square and no sticks with each
other. Removing one shared stick bre...

Q52. Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 8, R = 2, B = 6, L = 3, Centre = 41
Wheel 2: T = 10, R = 4, B = 5, L = 2, Centre = 65
Wheel 3: T = 12, R = 3, B = 9, L = 5, Centre = ?
(a) 85
(b) 81
(c) 90
(d) 78
Answer: **(A)** | *Explanation:* 85. Rule: Centre = $(T-L)^2 + (B-R)^2$. Wheel 1: $(8-3)^2+(6-2)^2 = 25+16 = 41$ ✓. Wheel 2: $(10-2)^2+(5-4)^2 =$
$64+1 = 65$ ✓. Wheel 3: $(12-5)^2+(9-3)^2$ =...

Q53. Matrix logic. Find the missing number.
3
7
13
7
12
19
13

19
?
(a) 25
(b) 27
(c) 29
(d) 24
Answer: **(B)** | *Explanation:* 27. Rule: cell(r,c) = $r^2+c^2+r\times c$ (symmetric in r and c, so the matrix itself is symmetric about its diagonal
— note how cell(1,2)=cell(2,1)=...

Q54. Pattern. Find the next term: 3, 9, 19, 33, 51, ?
(a) 73
(b) 69
(c) 77
(d) 71
Answer: **(A)** | *Explanation:* 73. Each term equals $2n^2+1$ for n=1,2,3,...: 3,9,19,33,51; next (n=6): 2(36)+1 = 73. Section 6: Grammar...

SECTION 6: English Grammar (10 min)

Q55. If the migration _____ properly tested, the system _____ experiencing intermittent failures right now.
(a) was, would not be
(b) had been, would not be
(c) were, will not be
(d) has been, would not have been
Answer: **(B)** | *Explanation:* A mixed conditional: "had been" (hypothetical past) with "would not be" (present result)....

Q56. The lead engineer said that the deployment _____ take place the following week, as the testing _____ not yet been completed at that time.
(a) would, had
(b) will, has
(c) would, has
(d) will, had
Answer: **(A)** | *Explanation:* Reported speech backshift: "will"→"would", "has"→"had"....

Q57. The team had to account _____ the discrepancy, which they eventually traced _____ a configuration error introduced _____ mistake during the last deployment.
(a) for, to, by
(b) of, to, in
(c) with, into, by
(d) for, into, with
Answer: **(A)** | *Explanation:* "account for" ... "traced to" ... "by mistake"....

Q58. Error spotting. "The new policy requires all employees to attending mandatory security training (A) / within thirty days of joining, (B) / failing which their system access (C) / will be automatically suspended. (D)"
(a) A

(b) B
(c) C
(d) D
Answer: **(A)** | *Explanation:* "requires...to attending" should be "to attend" (infinitive, not gerund) — the error is in part (A)....

Q59. _____ United Nations was founded after _____ World War II, and _____ UN Security Council now includes _____ five permanent members.
(a) The, the, the, the
(b) -, the, the, -
(c) The, -, the, the
(d) The, the, -, the
Answer: **(C)** | *Explanation:* "The United Nations" needs "the"; "World War II" conventionally takes no article; "the UN Security
Council" needs "the"; "the five permanen...

SECTION 7: English Writing (10 min)

Q60. Write a short technical essay/paragraph of 150–180 words.
Your team is choosing between two caching strategies for a high-traffic API: a write-through cache (data written to cache and database simultaneously) versus a write-back cache (data written to cache first, then asynchronously to the database). Explain the key trade-offs between these approaches in terms of data consistency and performance, and recommend which strategy best suits (a) a financial transactions API, and (b) a social media "like" counter — justifying each recommendation briefly.
Evaluation rubric (10 marks): Technical accuracy of the trade-off explanation (3) | Soundness of both context-specific recommendations (3) | Structure and coherence (2) | Grammar and vocabulary (1) | Word limit adherence (1)
Answer: **(N/A)** | *Explanation:* Evaluation Rubric (10 Marks): Architecture accuracy (3) | Trade-off comparison (3) | Recommendation logic (2) | Vocabulary (1) | Word limit ...

Mock Exam Paper 06: Infosys System Engineer ITM Mock Paper 6

Source File: `Infosys_SE_Mock_Paper_6.pdf`
Total Questions: 60 Questions | **Total Marks:** 69 Marks (59 MCQs + 10 Marks Writing)

SECTION 1: Reasoning Ability (25 min)

Q01. Statements: Some bugs are features. All features are releases. No release is stable. All patches are stable.
Conclusions: I. Some bugs are not stable. II. No patch is a feature. III. Some releases are bugs. IV. Some stable things are bugs.
(a) I and II only
(b) II and III only
(c) I, II and III only
(d) All follow
Answer: **(C)** | *Explanation:* Some bugs are features, features are releases, and no release is stable — so those

bugs are not stable
(I). Patches⊆stable and features∩sta...

Q02. Data Sufficiency. Is the product a × b positive?
I. a + b > 0.
II. a − b > 0.
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Even both statements together are not sufficient
Answer: **(D)** | *Explanation:* a=3, b=1 satisfies both statements and gives a positive product; a=3, b=−1 also satisfies both (a+b=2>0,
a−b=4>0) but gives a negative prod...

Q03. Data Sufficiency. Is the positive integer n a perfect square?
I. n has exactly 3 positive divisors.
II. n is divisible by 9.
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Even both statements together are not sufficient
Answer: **(A)** | *Explanation:* I: a number with exactly 3 divisors must be the square of a prime (p^2 has divisors 1, p, p^2), so n is a
perfect square — sufficient. II: n=...

Q04. In a code, the 1st letter of a word is shifted 1 place forward, the 2nd letter 2 places forward, the 3rd letter 3 places forward, and so on. Using this rule, CODE is coded as DQGI. What is the code for LOGIC?
(a) MQJMH
(b) MPJLH
(c) MQJNH
(d) MQIMH
Answer: **(A)** | *Explanation:* L+1=M, O+2=Q, G+3=J, I+4=M, C+5=H, giving MQJMH (verified: the same rule turns CODE into DQGI)....

Q05. In a code, the value of a word = (sum of the alphabetical positions of its letters × number of letters) − (number of vowels in the word). Given that CAB has the value 17, what is the value of DIGIT?
(a) 243
(b) 245
(c) 241
(d) 247
Answer: **(A)** | *Explanation:* DIGIT: positions 4+9+7+9+20 = 49; 49×5 = 245; vowels (I, I) = 2; 245−2 = 243 (check: CAB = 6×3−1 = 17
✓)....

Q06. A solid cube of side 7 units is painted on all faces and cut into 1-unit cubes. How many small cubes have at most 1 painted face?
(a) 275
(b) 125

(c) 150
(d) 335
Answer: **(A)** | *Explanation:* 0 painted faces (interior) = 5^3 = 125; exactly 1 painted face = 6×5^2 = 150; total = 275....

Q07. How many rectangles that are NOT squares can be found in a 4 × 5 grid of unit squares?
(a) 110
(b) 150
(c) 100
(d) 120
Answer: **(A)** | *Explanation:* Total rectangles = C(5,2)×C(6,2) = 10×15 = 150. Squares = 20+12+6+2 = 40. Non-square rectangles =
150−40 = 110....

Q08. If P # Q means P is the mother of Q, P $ Q means P is the brother of Q, and P @ Q means P is the wife of Q, then in the expression A $ B # C @ D, how is A related to D?
(a) Wife's maternal uncle (uncle-in-law)
(b) Brother-in-law
(c) Father-in-law
(d) Maternal uncle
Answer: **(A)** | *Explanation:* A is the brother of B, and B is the mother of C, so A is C's maternal uncle. C is the wife of D, so A is D's
wife's maternal uncle — an unc...

Q09. A is 20 km due North of B. C is 15 km due East of B. What is the bearing of A from C (measured clockwise
from North, to the nearest degree)?
(a) 323°
(b) 307°
(c) 037°
(d) 217°
Answer: **(A)** | *Explanation:* Taking East and North as axes, A is at (0,20), B at (0,0), C at (15,0). The vector from C to A is (−15, 20):
36.87° west of north, giving a...

Q10. Five people A, B, C, D and E stand in a queue. A is ahead of B but behind C. D stands immediately behind B. E is ahead of C. Who stands third from the front?
(a) A
(b) B
(c) C
(d) E
Answer: **(A)** | *Explanation:* E is ahead of C, C ahead of A, A ahead of B, and D immediately behind B: E, C, A, B, D. Third from the
front is A....

Q11. Find the next term: 3, 5, 11, 29, 83, ?
(a) 245
(b) 247

(c) 239
(d) 251
Answer: **(A)** | *Explanation:* Differences are 2, 6, 18, 54 (each ×3); the next difference is 162. 83+162 = 245....

Q12. Find the next term: 1, 4, 27, 16, 125, 36, ?
(a) 343
(b) 49
(c) 216
(d) 64
Answer: **(A)** | *Explanation:* Odd positions are cubes (1^3, 3^3, 5^3) and even positions are squares (2^2, 4^2, 6^2). The 7th term is 7^3 =
343....

Q13. Find the next group: ACF, GIL, MOR, ?
(a) SUX
(b) SVY
(c) TVY
(d) SUW
Answer: **(A)** | *Explanation:* Each group has three letters with gaps +2, +3 (A→C→F, G→I→L, M→O→R), and each group starts 6
letters after the previous one. The next group...

Q14. A square sheet of paper is folded in half three times (each fold halves the previous folded shape). A single hole is then punched through all layers. When the sheet is fully unfolded, how many holes does it have?
(a) 8
(b) 4
(c) 6
(d) 16
Answer: **(A)** | *Explanation:* Each fold doubles the layers: 2^3 = 8 layers, so one punch makes 8 holes....

Q15. Data Sufficiency. How is P related to Q?
I. P is the daughter of R.
II. R's brother is Q's father.
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Even both statements together are not sufficient
Answer: **(C)** | *Explanation:* I alone gives only P's parent; II alone links R and Q's father without involving P. Together: P is the
daughter of R, and R is the sibling ...

SECTION 2: Technical Ability (Quantitative) (35 min)

Q16. In how many ways can 8 people be seated around a circular table so that two particular people are NOT seated next to each other?
(a) 3600
(b) 5040

(c) 1440
(d) 4320
Answer: **(A)** | *Explanation:* Total circular arrangements = 7! = 5040. Arrangements with the two people adjacent = 2×6! = 1440.
Not adjacent = 5040−1440 = 3600....

Q17. How many 4-digit numbers (1000–9999) with all distinct digits are divisible by 5?
(a) 952
(b) 1008
(c) 896
(d) 504
Answer: **(A)** | *Explanation:* Last digit 0: 9×8×7 = 504. Last digit 5: the first digit has 8 choices (not 0 or 5), then 8 and 7 for the
middle: 8×8×7 = 448. Total = 504+...

Q18. Three fair dice are rolled. Given that at least one die shows a 6, what is the probability that the sum of the three dice is 12?
(a) 15/91
(b) 25/216
(c) 1/8
(d) 10/91
Answer: **(A)** | *Explanation:* Outcomes with at least one 6: $6^3-5^3 = 91$. Of these, the sum is 12 in 15 outcomes (permutations of
{6,5,1}: 6; {6,4,2}: 6; {6,3,3}: 3). Prob...

Q19. A disease affects 1% of a population. A test detects the disease in 95% of those who have it, but also gives a false positive for 10% of those who do not. If a randomly chosen person tests positive, what is the probability that they actually have the disease?
(a) 19/217
(b) 19/100
(c) 95/1000
(d) 1/11
Answer: **(A)** | *Explanation:* P(disease and positive) = 0.01×0.95 = 0.0095; P(no disease and positive) = 0.99×0.10 = 0.099.
P(disease | positive) = 0.0095/0.1085 = 19/21...

Q20. A merchant sells two articles at ₹1,200 each. On one he gains 25%, on the other he loses 20%. What is his overall result (approximately)?
(a) 2.4% loss
(b) 2.4% profit
(c) No gain, no loss
(d) 5% loss
Answer: **(A)** | *Explanation:* CP_1 = 1200/1.25 = 960; CP_2 = 1200/0.80 = 1500. Total CP = 2460, total SP = 2400, loss = 60. Loss % =
60/2460 ≈ 2.4% loss....

Q21. A is twice as efficient as B, and B is three times as efficient as C. Working together, all three complete a job in 9 days. In how many days can C alone complete it?

(a) 90 days
(b) 81 days
(c) 60 days
(d) 72 days
Answer: **(A)** | *Explanation:* Let C = 1 unit/day, B = 3, A = 6, so together 10 units/day. In 9 days the work is 90 units, so C alone
needs 90/1 = 90 days....

Q22. A 180 m long train passes a man walking at 6 km/h in the same direction in 12 seconds. How long will this train take to completely cross a 420 m long platform?
(a) 36 s
(b) 30 s
(c) 40 s
(d) 42 s
Answer: **(A)** | *Explanation:* Relative speed = 180/12 = 15 m/s = 54 km/h; the train's speed = 54+6 = 60 km/h = 50/3 m/s. Time to
cross the platform = (180+420)/(50/3) = ...

Q23. A man spends 40% of his salary on food, 25% of the remainder on rent, and 20% of what is then left on travel. He is left with ₹3,600. What is his salary?
(a) ₹10,000
(b) ₹12,000
(c) ₹9,000
(d) ₹11,000
Answer: **(A)** | *Explanation:* After food: 60%. After rent (25% of 60% = 15%): 45%. After travel (20% of 45% = 9%): 36%. 36% of
salary = 3600, so the salary = ₹10,000....

Q24. A loan is repaid in two equal annual installments of ₹2,420 each, with interest at 10% per annum compounded annually (first installment due at the end of year 1). What was the loan amount?
(a) ₹4,200
(b) ₹4,400
(c) ₹4,000
(d) ₹4,840
Answer: **(A)** | *Explanation:* Loan = $2420/1.1 + 2420/1.1^2 = 2200 + 2000 = ₹4,200$....

Q25. In what ratio must milk-water mixtures of strength 3:2 and 5:1 (milk:water) be mixed to obtain a mixture of strength 2:1?
(a) 5:2
(b) 2:5
(c) 3:2
(d) 1:3
Read the passage and answer Q26–30.
When a measure becomes a target, it ceases to be a good measure — a maxim known as Goodhart's law that engineering organisations rediscover with remarkable regularity. A team told that its performance will be judged by the number of closed tickets quickly learns to split large tasks into many trivial ones; a group rewarded for code coverage writes tests that execute lines without asserting anything meaningful. In neither case is the behaviour necessarily cynical. People respond rationally to the incentives they

perceive, and the metric, once elevated to a goal, simply redirects that rationality away from the outcome it was intended to track. The remedy is not to abandon measurement, which would leave managers navigating by anecdote, but to treat any single metric as a symptom rather than a verdict, to pair it with a counter-metric that penalises its obvious abuse, and to revisit the pairing as teams learn how to game it. Even then, the arrangement is only ever provisionally sound, since every safeguard becomes, in time, another target.

Answer: **(A)** | *Explanation:* Milk fractions: 3/5, 5/6, target 2/3. By alligation, the amounts are in the ratio (5/6−2/3) : (2/3−3/5) =

1/6 : 1/15 = 5:2.

Section 3: Ver...

SECTION 3: Verbal Ability (20 min)

Reading Comprehension Passage (Q26–Q30):

```
Read the passage and answer Q26–30.
When a measure becomes a target, it ceases to be a good measure — a maxim known as Goodhart's law
that engineering organisations rediscover with remarkable regularity. A team told that its
performance
will be judged by the number of closed tickets quickly learns to split large tasks into many
trivial ones; a
group rewarded for code coverage writes tests that execute lines without asserting anything
meaningful.
In neither case is the behaviour necessarily cynical. People respond rationally to the incentives
they
perceive, and the metric, once elevated to a goal, simply redirects that rationality away from the
outcome
it was intended to track. The remedy is not to abandon measurement, which would leave managers
navigating by anecdote, but to treat any single metric as a symptom rather than a verdict, to pair
it with
a counter-metric that penalises its obvious abuse, and to revisit the pairing as teams learn how
to game
it. Even then, the arrangement is only ever provisionally sound, since every safeguard becomes, in
time,
another target.

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Q26. The passage's main point is that:

(a) Engineering metrics should be abolished

(b) Metrics that become targets get gamed, so they should be paired with counter-metrics and revisited, not dropped

(c) Developers deliberately deceive their managers

(d) Code coverage is the best available metric

Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 6, option (B) is directly supported and established by the author's primary thesis ...

Q27. The author's view of the gaming behaviour described is that it is:

(a) Usually a sign of dishonesty

(b) Not necessarily cynical, but a rational response to perceived incentives

(c) Impossible to prevent
(d) Limited to junior engineers
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 6, option (B) is directly supported and established by the author's primary thesis ...

Q28. Treating a metric "as a symptom rather than a verdict" most nearly means:
(a) Ignoring the metric entirely
(b) Using the metric as a signal that prompts investigation rather than a final judgement of performance
(c) Replacing the metric every quarter
(d) Publishing the metric to all staff
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 6, option (B) is directly supported and established by the author's primary thesis ...

Q29. According to the passage, why would abandoning measurement be undesirable?
(a) It would reduce team morale
(b) It would make teams ignore tickets
(c) Managers would be left relying on anecdote
(d) It would violate Goodhart's law
Answer: **(C)** | *Explanation:* According to the reading comprehension passage for Paper 6, option (C) is directly supported and established by the author's primary thesis ...

Q30. The final sentence implies that:
(a) Even a counter-metric can eventually be gamed, so the arrangement needs periodic review
(b) Counter-metrics are worse than the original metrics
(c) Safeguards should never be introduced
(d) Teams eventually stop responding to incentives
Sentence Correction (Q31–35).
Answer: **(A)** | *Explanation:* According to the reading comprehension passage for Paper 6, option (A) is directly supported and established by the author's primary thesis ...

Q31. Neither the project manager nor the developers was informed about the change.
(a) No improvement
(b) were informed
(c) is informed
(d) has been informed
Answer: **(B)** | *Explanation:* With "neither…nor", the verb agrees with the nearer subject (developers → were)....

Q32. Walking into the server room, the temperature seemed unusually high to the technician.
(a) No improvement
(b) the technician found the temperature unusually high
(c) the temperature seemed to
the technician unusually high
(d) it was unusually high to the technician
Answer: **(B)** | *Explanation:* The opening participle must modify a person; "the technician found the temperature…" fixes the
dangling modifier....

Q33. The new framework is faster, more secure, and it is easy to maintain.
(a) No improvement

(b) and easier to maintain
(c) and also is easy for maintenance
(d) and maintaining it is
easy
Answer: **(B)** | *Explanation:* The adjectives must stay parallel: faster, more secure, and easier to maintain....

Q34. By the time the patch was released, the attackers already exploited the vulnerability.
(a) No improvement
(b) had already exploited
(c) have already exploited
(d) were already exploiting
Answer: **(B)** | *Explanation:* An action completed before another past action needs the past perfect: "had already exploited"....

Q35. The reason why the build failed is because of a missing dependency.
(a) No improvement
(b) The reason the build failed was because a dependency was missing
(c) The build
failed because a dependency was missing
(d) The reason why the build failed was because of a missing
dependency
Para-jumbles (Q36–38).
Answer: **(C)** | *Explanation:* "The reason… is because" is redundant; "The build failed because a dependency was missing" is the
clean form....

Q36. P: Consequently, many teams now treat flaky tests as defects in their own right.
Q: A test that passes and fails intermittently without any code change erodes trust in the whole suite.
R: Developers begin to ignore failures, assuming they are noise.
S: Real regressions then slip through unnoticed.
(a) QRSP
(b) QSRP
(c) RQSP
(d) QRPS
Answer: **(A)** | *Explanation:* Sentence sequence: Q→R→S→P.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q37. P: Releasing to every user at once maximises speed but also the blast radius of any defect.
Q: Teams face a genuine tension between releasing quickly and releasing safely.
R: Releasing to a handful of users first limits risk but delays feedback.
S: A staged rollout, exposing a small and growing share of users, is the usual compromise.
(a) QPRS
(b) QRPS
(c) PQRS
(d) QPSR
Answer: **(A)** | *Explanation:* Sentence sequence: Q→P→R→S.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q38. P: Its adoption, however, was slow outside the research community.
Q: Public-key cryptography solved the problem of exchanging secret keys over an insecure channel.
R: Usable libraries and standards eventually removed this barrier.
S: Early implementations were notoriously difficult to use correctly.
(a) QSPR
(b) QPSR
(c) PQSR
(d) QPRS
Vocabulary, Synonyms and Antonyms (Q39–45).
Answer: **(B)** | *Explanation:* Sentence sequence: Q→P→S→R.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q39. The documentation was admirably pellucid, leaving no step open to misreading. "Pellucid" means:
(a) lengthy
(b) transparently clear
(c) outdated
(d) technical
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q40. The client's mercurial moods made planning difficult. "Mercurial" means:
(a) steady
(b) unpredictably changeable
(c) hostile
(d) cheerful
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q41. Fill in the blank: The startup's growth was _____; revenue tripled in a single quarter.
(a) meteoric
(b) gradual
(c) stagnant
(d) negligible
Answer: **(A)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (A) is the exact synonym/antonym/definition required in this context....

Q42. Synonym of EQUIVOCATE:
(a) speak ambiguously to avoid commitment
(b) confess openly
(c) argue loudly
(d) agree readily
Answer: **(A)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (A) is the exact synonym/antonym/definition required in this context....

Q43. Synonym of PERNICIOUS:
(a) harmless
(b) subtly or gravely harmful
(c) trivial

(d) beneficial
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q44. Antonym of BENEVOLENT:
(a) kind
(b) malevolent
(c) generous
(d) charitable
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q45. Antonym of EXACERBATE:
(a) worsen
(b) aggravate
(c) alleviate
(d) intensify
Answer: **(C)** | *Explanation:* Section 4: Pseudo Code...

SECTION 4: Pseudo Code (10 min)

Q46. What is printed? (/ is integer division, % is modulo)
```
int f(int n) {
if (n <= 1) return n;
if (n % 2 == 0) return f(n / 2);
return f(n - 1) + f(n + 1);
}
print(f(11))
```
(a) 5
(b) 4
(c) 6
(d) 3
Answer: **(A)** | *Explanation:* 5. Building up: f(1)=1, f(2)=f(1)=1, f(4)=f(2)=1, f(3)=f(2)+f(4)=2, f(6)=f(3)=2, f(5)=f(4)+f(6)=3,
f(10)=f(5)=3, f(12)=f(6)=2. So f(11)=f(1...

Q47. What is printed? (^=XOR, >>=right shift, &=AND)
```
n = 13
g = n ^ (n >> 1)
print(g + (g & 5))
```
(a) 12
(b) 11
(c) 16
(d) 14
Answer: **(A)** | *Explanation:* 12. 13 = 1101_2; 13>>1 = 6 = 0110_2; g = 1101 ^ 0110 = 1011_2 = 11. g & 5 = 1011 & 0101 = 0001 = 1. g + 1
= 12....

Q48. What is printed?
s = 0
for i = 1 to 5
for j = 1 to 5
if ((i + j) % 4 == 0) continue
if (j > i + 1) break
s = s + i * j
end for
end for
print(s)
(a) 142
(b) 138
(c) 146
(d) 150
Answer: **(A)** | *Explanation:* 142. Tracing the loops with the continue (when i+j is a multiple of 4) and break (when j > i+1)
gives a total of 142 (confirmed by code)....

Q49. The following sorts a stack using a temporary stack (the top of a stack is its last element). How many times is an element moved from tmp back to st?
st = [3, 1, 4, 2]; tmp = []; moves = 0
while (st is not empty)
t = pop(st)
while (tmp is not empty and top(tmp) > t)
push(st, pop(tmp))
moves = moves + 1
end while
push(tmp, t)
end while
print(moves)
(a) 3
(b) 2
(c) 4
(d) 5
Answer: **(A)** | *Explanation:* 3. Tracing: pop 2 → tmp [2]; pop 4 → tmp [2,4]; pop 1 → move 4 and 2 back (2 moves) → tmp [1];
pop 2 → tmp [1,2]; pop 4 → tmp [1,2,4]; pop ...

Q50. What is printed? (dynamic programming for the minimum number of coins; INF is a very large number)
coins = [1, 3, 4]
dp[0] = 0; dp[1..6] = INF
for i = 1 to 6
for each c in coins
if (i >= c)
dp[i] = min(dp[i], dp[i - c] + 1)
end if

end for
end for
print(dp[6])
(a) 2
(b) 3
(c) 4
(d) 6
Answer: **(A)** | *Explanation:* 2. dp = [0, 1, 2, 1, 1, 2, 2], so dp[6] = 2 (3+3). A greedy approach (4+1+1) would give 3, which is why
the DP matters.

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SECTION 5: Numeric Puzzle Solving (10 min)

Q51. Matchstick. A 2 × 3 grid of unit squares (2 rows, 3 columns) is built from 17 matchsticks and contains 8 squares in total (six 1×1 and two 2×2). What is the minimum number of sticks that must be removed so that no complete square of any size remains?
(a) 3
(b) 4
(c) 5
(d) 6
Answer: **(B)** | *Explanation:* 4. Exhaustive computer search over all 17 sticks and all 8 squares shows that no set of 3 sticks breaks
every square, while a set of 4 does...

Q52. Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 4, R = 1, B = 3, L = 2, Centre = 40
Wheel 2: T = 6, R = 2, B = 2, L = 3, Centre = 39
Wheel 3: T = 5, R = 3, B = 4, L = 1, Centre = ?
(a) 65
(b) 61
(c) 69
(d) 57
Answer: **(A)** | *Explanation:* 65. Rule: Centre = $(T+B)^2 - (R+L)^2$. Wheel 1: $7^2-3^2 = 40$ ✓. Wheel 2: $8^2-5^2 = 39$ ✓. Wheel 3: 9^2-4^2 =
81−16 = 65....

Q53. Matrix logic. Find the missing number.
5
7
18
6
11
25
8
9

?
(a) 55
(b) 73
(c) 64
(d) 47
Answer: **(A)** | *Explanation:* 55. Rule: third number = (first)^2 − second. Row 1: 25−7 = 18 ✓. Row 2: 36−11 = 25 ✓. Row 3: 64−9 =
55....

Q54. Pattern. Find the next term: 2, 10, 30, 68, 130, ?
(a) 222
(b) 216
(c) 210
(d) 228
Answer: **(A)** | *Explanation:* 222. Each term is n^3+n: 2, 10, 30, 68, 130; for n=6: 216+6 = 222.
Section 6: Grammar...

SECTION 6: English Grammar (10 min)

Q55. The quality of the algorithms, along with the efficiency of the data structures, _____ determined the system's throughput.
(a) have
(b) has
(c) were
(d) are
Answer: **(B)** | *Explanation:* The true subject is "the quality" (singular); "along with..." does not make it plural → has....

Q56. No sooner _____ the alert been raised than the on-call engineer responded.
(a) had
(b) has
(c) did
(d) was
Answer: **(A)** | *Explanation:* "No sooner" takes the past perfect with inversion: "No sooner had the alert been raised than..."....

Q57. He is oblivious _____ the risks, though he was warned _____ them repeatedly, and he insists _____ proceeding.
(a) of, about, in
(b) to, of, on
(c) to, for, in
(d) with, of, on
Answer: **(B)** | *Explanation:* oblivious to, warned of, insists on....

Q58. Error spotting. "Despite of the repeated warnings (A) / the team continued to deploy (B) / changes directly to production, (C) / which was clearly risky. (D)"
(a) A
(b) B

(c) C
(d) D
Answer: **(A)** | *Explanation:* "Despite of" is incorrect; it should be "Despite" or "In spite of"....

Q59. _____ European Union has adopted _____ unique approach to regulating _____ artificial intelligence, and _____ honest debate about its merits continues.
(a) The, a, -, an
(b) A, a, the, an
(c) The, an, -, a
(d) The, a, the, a
Answer: **(A)** | *Explanation:* The European Union (proper institution), a unique (starts with /j/ sound), artificial intelligence
(general concept, no article), an honest...

SECTION 7: English Writing (10 min)

Q60. Write an email of 150–180 words.
While reviewing a client's production application, you discover a security vulnerability that could expose customer data if exploited. Write an email to the client's CTO that (i) informs them of the issue clearly but without causing panic or assigning blame, (ii) states the immediate mitigation you have already applied, (iii) outlines the permanent fix you propose, and (iv) requests an urgent meeting.
Evaluation rubric (10 marks): Subject line, format and tone (2) | Clear, calm explanation of the issue and risk (2) | Mitigation and permanent fix described (3) | Clear call to action (1) | Grammar, vocabulary and word limit (2)
Answer: **(N/A)** | *Explanation:* Evaluation Rubric (10 Marks): Subject & format (2) | Neutral risk framing (3) | Immediate mitigation & plan (3) | Tone & vocabulary (1) | Wo...

Mock Exam Paper 07: Infosys System Engineer ITM Mock Paper 7

Source File: `Infosys_SE_Mock_Paper_7.pdf`
Total Questions: 60 Questions | **Total Marks:** 69 Marks (59 MCQs + 10 Marks Writing)

SECTION 1: Reasoning Ability (25 min)

Q01. Statements: Some servers are clients. All clients are nodes. No node is offline. Some offline things are routers.
Conclusions: I. Some servers are not offline. II. Some routers are not nodes. III. No client is a router. IV. Some nodes are servers.
(a) I and II only
(b) I, II and IV only
(c) I, II, III and IV
(d) II, III and IV only
Answer: **(B)** | *Explanation:* Some servers are clients and clients are nodes, which are never offline, so those servers are not offline
(I). Some offline things are rout...

Q02. Data Sufficiency. What is the value of x + y?
I. $x^2 + 2xy + y^2 = 49$
II. x and y are both positive.
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Even both statements together are not sufficient
Answer: **(C)** | *Explanation:* I gives $(x + y)^2 = 49$, so x + y = 7 or −7. II alone gives no value. Together, x + y must be positive, so x + y =
7....

Q03. Data Sufficiency. Is the average of the set S greater than 10?
I. The median of S is 12.
II. S contains exactly 5 positive integers.
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Even both statements together are not sufficient
Answer: **(D)** | *Explanation:* A set such as {1, 1, 12, 12, 12} has median 12 and 5 positive integers but an average of 7.6, while {12,
12, 12, 13, 100} has an average ab...

Q04. In a code, each letter is replaced by its opposite letter in the alphabet (A↔Z, B↔Y, C↔X, and so on), and the resulting word is then written in reverse order. Using this rule, CODE is coded as VWLX. What is the code for SYSTEM?
(a) NVGHBH
(b) HBHGVN
(c) NVGHBG
(d) MVGHBH
Answer: **(A)** | *Explanation:* S→H, Y→B, S→H, T→G, E→V, M→N gives HBHGVN, which reversed is NVGHBH (check:
CODE→XLWV→VWLX ✓)....

Q05. In a certain code, ROSE is written as 6821, CHAIR is written as 73456, and PREACH is written as 961473. How is SEARCH written in that code?
(a) 214673
(b) 214763
(c) 241673
(d) 214637
Questions 6 and 7 are based on the following information.
Eight people A, B, C, D, E, F, G and H sit around a circular table facing the centre. A sits opposite E. B sits immediately to the right of A. C sits second to the left of B. D sits immediately to the right of E. G sits opposite C. F sits second to the right of A.
Answer: **(A)** | *Explanation:* From the three words: R=6, O=8, S=2, E=1, C=7, H=3, A=4, I=5, P=9. SEARCH = S(2) E(1) A(4) R(6) C(7)
H(3) = 214673....

Q06. Who sits opposite B?
(a) D
(b) H
(c) F
(d) G
Answer: **(A)** | *Explanation:* Fix A at one seat and number the seats clockwise from A: A0, C1, H2, D3, E4, G5, F6, B7. (E is opposite A
at 4; B is to A's right at 7; C i...

Q07. Who sits third to the left of F?
(a) C
(b) A
(c) H
(d) B
Questions 8 and 9 are based on the following table.
Year
Revenue (₹ crore)
Expenditure (₹ crore)
2021
420
350
2022
510
400
2023
600
480
2024
660
550
2025
780
600

(Profit = Revenue − Expenditure.)
Answer: **(A)** | *Explanation:* F is at seat 6. Moving to the left (clockwise): first is B (7), second is A (0), third is C (1)....

Q08. Over the five years together, total profit is what percentage of total expenditure (to the nearest whole number)?
(a) 25%
(b) 24%
(c) 26%
(d) 28%
Answer: **(A)** | *Explanation:* Profits: 70, 110, 120, 110, 180, totalling 590. Total expenditure = 350+400+480+550+600 = 2380.
590/2380 ≈ 24.8%, which is 25%....

Q09. Between which two consecutive years was the percentage growth in revenue the lowest?
(a) 2021–2022
(b) 2022–2023
(c) 2023–2024
(d) 2024–2025
Answer: **(C)** | *Explanation:* Growth: 2021–22 = 21.4%, 2022–23 = 17.6%, 2023–24 = 10%, 2024–25 = 18.2%. The lowest is 2023–
2024....

Q10. A is the mother of B. B is the sister of C. C is the father of D. E is the only brother of A. How is E related to D?
(a) Paternal grand-uncle
(b) Maternal uncle
(c) Paternal uncle
(d) Grandfather
Answer: **(A)** | *Explanation:* A is the mother of B, and B is C's sister, so A is also C's mother. C is D's father, so A is D's paternal
grandmother. E is A's only brothe...

Q11. A ship sails 20 km on a bearing of 030°, and then 20 km on a bearing of 330°. What is its final position relative to the starting point?
(a) 20√3 km (about 34.6 km) due North
(b) 20 km due North
(c) 40 km due North
(d) 20√3 km due East
Answer: **(A)** | *Explanation:* Leg 1: East = 20 sin30° = 10, North = 20 cos30° = 10√3. Leg 2 (bearing 330°): East = −10, North =
+10√3. The East parts cancel; the North p...

Q12. Find the next term: 1, 3, 12, 60, 360, ?
(a) 2520
(b) 2160
(c) 1800
(d) 2400
Answer: **(A)** | *Explanation:* Each term is the previous term multiplied by 3, 4, 5, 6 and then 7: 360 × 7 = 2520....

Q13. Find the next term: 3, 8, 6, 14, 12, 26, ?
(a) 24
(b) 22
(c) 50
(d) 52
Answer: **(A)** | *Explanation:* Two interleaved series. Odd positions: 3, 6, 12, then ×2 gives 24. Even positions: 8, 14, 26 (each is
double the previous minus 2). The 7th...

Q14. The image of a wall clock in a plane mirror shows the time as 8:25. What is the actual time?
(a) 3:35
(b) 4:35

(c) 3:25
(d) 4:25
Answer: **(A)** | *Explanation:* Actual time = 11:60 − 8:25 = 3:35....

Q15. In a row of 40 students, A is 15th from the left and B is 18th from the right. How many students are between A and B?
(a) 7
(b) 6
(c) 8
(d) 9
Answer: **(A)** | *Explanation:* B is 18th from the right, so B is 40−18+1 = 23rd from the left. Students between A (15th) and B (23rd)
= 23−15−1 = 7.

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SECTION 2: Technical Ability (Quantitative) (35 min)

Q16. In how many ways can the letters of the word SUCCESS be arranged so that no two S's are adjacent?
(a) 120
(b) 60
(c) 240
(d) 420
Answer: **(A)** | *Explanation:* Arrange the non-S letters U, C, C, E first: 4!/2! = 12 ways. This creates 5 gaps, and the 3 S's go into 3
different gaps: C(5,3) = 10. Tota...

Q17. A fair coin is tossed 6 times. What is the probability that no two heads appear consecutively?
(a) 21/64
(b) 13/32
(c) 1/3
(d) 5/16
Answer: **(A)** | *Explanation:* The number of binary strings of length 6 with no two consecutive heads is the Fibonacci number 21,
out of 2^6 = 64. Probability = 21/64 (con...

Q18. A shopkeeper marks his goods 40% above cost, gives a 15% discount on the marked price, and also uses a weight of 950 g in place of 1 kg. What is his actual profit percentage (approximately)?
(a) 25.26%
(b) 19%
(c) 24%
(d) 26.3%
Answer: **(A)** | *Explanation:* Take cost per kg = 100. Marked price = 140; after a 15% discount the selling price for a claimed 1 kg is
119. The customer actually gets on...

Q19. A and B together can do a job in 12 days, B and C together in 15 days, and C and A together in 20 days. In how many days can A alone do the job?
(a) 30 days

(b) 24 days
(c) 20 days
(d) 40 days
Answer: **(A)** | *Explanation:* (A+B) + (B+C) + (C+A) = 1/12 + 1/15 + 1/20 = 12/60 = 1/5, so A+B+C = 1/10. A = 1/10 − (B+C) = 1/10 −
1/15 = 1/30, so A alone needs 30 days....

Q20. A boat takes 6 hours to go 24 km upstream and 28 km downstream, and 6½ hours to go 30 km upstream and 21 km downstream. What is the speed of the boat in still water?
(a) 10 km/h
(b) 8 km/h
(c) 12 km/h
(d) 9 km/h
Answer: **(A)** | *Explanation:* Let upstream speed = u and downstream speed = d. 24/u + 28/d = 6 and 30/u + 21/d = 6.5. Solving
gives u = 6 and d = 14 (check: 4 + 2 = 6 ✓,...

Q21. In an examination, 70% of the candidates passed in Mathematics, 60% passed in Science, and 15% failed in both subjects. What percentage of the candidates passed in both subjects?
(a) 45%
(b) 40%
(c) 50%
(d) 55%
Answer: **(A)** | *Explanation:* Passed in at least one subject = 100 − 15 = 85%. Both = 70 + 60 − 85 = 45%....

Q22. The difference between the compound interest (annual compounding) and the simple interest on a sum for 3 years at 20% per annum is ₹640. What is the sum?
(a) ₹5,000
(b) ₹4,000
(c) ₹6,000
(d) ₹5,500
Answer: **(A)** | *Explanation:* For 3 years at 20%, CI − SI = P(1.2^3 − 1 − 0.6) = 0.128P = 640, so P = ₹5,000....

Q23. What is the remainder when 2^{2026} is divided by 7?
(a) 2
(b) 1
(c) 4
(d) 3
Answer: **(A)** | *Explanation:* 2^3 = 8 ≡ 1 (mod 7), so powers of 2 repeat every 3. 2026 = 3×675 + 1, so 2^{2026} ≡ 2^1 = 2....

Q24. The present ages of A and B are in the ratio 5 : 3. After 4 years, the ratio of their ages will be 3 : 2. What is the sum of their present ages?
(a) 32 years
(b) 36 years
(c) 40 years
(d) 28 years
Answer: **(A)** | *Explanation:* Let the ages be 5x and 3x. (5x+4)/(3x+4) = 3/2 gives 10x + 8 = 9x + 12, so x = 4.

The ages are 20 and 12,
and the sum is 32....

Q25. Three bells ring at intervals of 12, 15 and 20 minutes. They all ring together at 9:00 AM. How many times in all (including 9:00 AM) will they ring together between 9:00 AM and 3:00 PM, both times included?
(a) 7
(b) 6
(c) 8
(d) 5
Read the passage and answer Q26–30.
In distributed systems, a network call that times out leaves the caller in genuine ignorance: the request may have been lost before reaching the server, executed successfully with the reply lost on the way back, or still be in flight. Retrying is the natural response, but retrying a non-idempotent operation — charging a card, say — risks performing it twice. The standard remedy is to make the operation idempotent by attaching a unique request identifier, so that the server can recognise a duplicate and return the original result instead of re-executing. This shifts the burden rather than removing it: the server must now durably remember identifiers, and must decide how long to remember them, since storing them forever is costly while discarding them too early reopens the window for duplicates. Engineers who treat idempotency as a mere implementation detail therefore tend to discover it as an incident, whereas those who treat it as part of an operation's contract design it in from the start.
Answer: **(A)** | *Explanation:* LCM(12, 15, 20) = 60 minutes. In 6 hours (360 minutes) the bells ring together at 0, 60, 120, 180, 240,
300 and 360 minutes: 7 times.
Sect...

SECTION 3: Verbal Ability (20 min)

Reading Comprehension Passage (Q26–Q30):

```
Read the passage and answer Q26–30.
In distributed systems, a network call that times out leaves the caller in genuine ignorance: the
request
may have been lost before reaching the server, executed successfully with the reply lost on the
way back,
or still be in flight. Retrying is the natural response, but retrying a non-idempotent operation —
charging
a card, say — risks performing it twice. The standard remedy is to make the operation idempotent
by
attaching a unique request identifier, so that the server can recognise a duplicate and return the
original
result instead of re-executing. This shifts the burden rather than removing it: the server must
now durably
remember identifiers, and must decide how long to remember them, since storing them forever is
costly
while discarding them too early reopens the window for duplicates. Engineers who treat idempotency
as
a mere implementation detail therefore tend to discover it as an incident, whereas those who treat
it as
part of an operation's contract design it in from the start.
```

Q26. The passage mainly argues that:
(a) Network timeouts can be eliminated with better hardware
(b) Safe retries need idempotency, which has its own costs and is best designed in from the start
(c) Card payments should never be retried
(d) Servers should discard request identifiers immediately
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 7, option (B) is directly supported and established by the author's primary thesis ...

Q27. "Genuine ignorance" in the first sentence refers to the fact that:
(a) the caller does not know the server's address
(b) the server refuses to reply
(c) the caller cannot tell whether the request was executed
(d) the network is always unreliable
Answer: **(C)** | *Explanation:* According to the reading comprehension passage for Paper 7, option (C) is directly supported and established by the author's primary thesis ...

Q28. The author says the identifier approach "shifts the burden rather than removing it" because:
(a) the caller must now generate more traffic
(b) the server must store identifiers and decide how long to keep them
(c) the identifiers make the operation non-idempotent
(d) duplicates become impossible to detect
Answer: **(B)** | *Explanation:* According to the reading comprehension passage for Paper 7, option (B) is directly supported and established by the author's primary thesis ...

Q29. According to the passage, what is the risk of discarding identifiers too early?
(a) A duplicate request may be executed a second time
(b) The server will run out of storage
(c) The caller will stop retrying
(d) The original result will be corrupted
Answer: **(A)** | *Explanation:* According to the reading comprehension passage for Paper 7, option (A) is directly supported and established by the author's primary thesis ...

Q30. Which inference is best supported by the last sentence?
(a) Idempotency is only needed in payment systems
(b) Incidents are caused only by implementation details
(c) Teams that include idempotency in an operation's contract are less likely to meet it first as an outage
(d) Contracts should not describe retry behaviour

Critical Reasoning (Q31–33).
Answer: **(C)** | *Explanation:* According to the reading comprehension passage for Paper 7, option (C) is directly supported and established by the author's primary thesis ...

Q31. After a website adopted a new CDN, its page-load time fell by 30%, and user retention rose over the same period. The team concludes that the CDN caused the rise in retention. Which of the following is an assumption on which this conclusion depends?
(a) Retention is determined only by page-load time
(b) No other factor that affects retention changed significantly over the same period
(c) All CDNs improve retention
(d) A 30% fall in load time is always considered large

Answer: **(B)** | *Explanation:* The conclusion that the CDN caused the retention rise only holds if nothing else that affects retention
changed over the same period....

Q32. Since a company made code reviews mandatory, production defects have fallen by 40%. The company concludes that code reviews reduce defects. Which of the following, if true, most weakens this conclusion?
(a) Each review takes about two hours a week
(b) In the same period the company halved its release frequency and cut new-feature work sharply
(c) Developers say they like the review process
(d) The defect-tracking system was not changed
Answer: **(B)** | *Explanation:* Halving releases and cutting feature work gives an alternative explanation for fewer defects, so it
weakens the claim that reviews were the...

Q33. A start-up claims its new caching layer will cut its database costs. Which of the following, if true, most strengthens the claim?
(a) In a pilot, 85% of requests were served from the cache, and every cache hit avoids a billable database query
(b) The caching layer was designed by a well-known engineer
(c) The start-up's database costs rose last year
(d) The caching layer is built on open-source software
Sentence Correction (Q34–36).
Answer: **(A)** | *Explanation:* A high cache-hit rate, with each hit avoiding a billable query, directly links the cache to lower
database costs....

Q34. The data, which was collected over three years, shows that fewer people has adopted the system than expected.
(a) No improvement
(b) fewer people have adopted
(c) less people have adopted
(d) fewer people is
adopting
Answer: **(B)** | *Explanation:* "Fewer" is used with countable nouns (people), and the plural subject "people" takes "have"....

Q35. The throughput of the new cluster is much higher than the old one.
(a) No improvement
(b) than that of the old one
(c) than the old one's
(d) from the old one
Answer: **(B)** | *Explanation:* Compare like with like: the throughput of the new cluster with that of the old one....

Q36. I wish the system was more reliable than it is now.
(a) No improvement
(b) were
(c) is
(d) would be

Para-jumbles (Q37–39).
Answer: **(B)** | *Explanation:* After "wish", a present unreal condition takes the subjunctive "were"....

Q37. P: This is why eventually consistent replicas can briefly serve stale reads.
Q: In a replicated database, a write is first applied to one node and then propagated to the others.
R: This window may be small, but it is never zero.
S: During propagation, different nodes hold different versions of the same record.
(a) QRSP
(b) QSRP
(c) SQRP
(d) QSPR
Answer: **(B)** | *Explanation:* Sentence sequence: Q→S→R→P.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q38. P: Hash tables achieve average constant-time lookup.
Q: Collisions, however, are inevitable.
R: Chaining and open addressing are the two standard ways to resolve them.
S: If many keys land in the same bucket, performance degrades towards linear time.
(a) PQRS
(b) PQSR
(c) QPSR
(d) PSQR
Answer: **(B)** | *Explanation:* Sentence sequence: P→Q→S→R.. Coherent narrative progression establishes logical flow from premise to supporting detail and conclusion....

Q39. P: That decision, in hindsight, saved the company millions.
Q: In 2019 the board chose to rewrite its payment platform rather than patch it.
R: Patching would have cost less that year.
S: The rewrite also delayed several product launches.
(a) QRSP
(b) QSRP
(c) RQSP
(d) QSPR
Sentence Completion (Q40–41).
Answer: **(A)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (A) is the exact synonym/antonym/definition required in this context....

Q40. Although the prototype looked impressive in the demo, its performance under real load was _____.
(a) unimpeachable
(b) underwhelming
(c) exemplary
(d) gratifying
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q41. The engineer's explanation was so _____ that even the non-technical audience grasped the idea at once.
(a) convoluted

(b) lucid
(c) esoteric
(d) verbose
Vocabulary, Synonyms and Antonyms (Q42–45).
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q42. Synonym of ABSTRUSE:
(a) obvious
(b) difficult to understand
(c) lengthy
(d) outdated
Answer: **(B)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (B) is the exact synonym/antonym/definition required in this context....

Q43. Synonym of SALIENT:
(a) minor
(b) hidden
(c) most noticeable or important
(d) rejected
Answer: **(C)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (C) is the exact synonym/antonym/definition required in this context....

Q44. Antonym of ARDUOUS:
(a) strenuous
(b) demanding
(c) effortless
(d) lengthy
Answer: **(C)** | *Explanation:* Standard vocabulary / grammatical definition establishes that option (C) is the exact synonym/antonym/definition required in this context....

Q45. Antonym of OBSEQUIOUS:
(a) servile
(b) fawning
(c) assertive
(d) flattering
Answer: **(C)** | *Explanation:* Infosys System Engineer |Mock Test
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Section 4: Pseudo Code...

SECTION 4: Pseudo Code (10 min)

Q46. What is printed? (Operands are evaluated strictly from left to right; x++ and x-- return the old value and then change x, while ++x and --x change x first and then return the new value.)

```
x = 4
y = x++ * 2 + ++x
z = --x + x-- * 2
print(x + y + z)
```

(a) 33
(b) 31
(c) 35
(d) 29
Answer: **(A)** | *Explanation:* 33. Line 2: x++ returns 4 (x becomes 5); ×2 = 8; ++x makes x = 6 and returns 6; y = 8 + 6 = 14, x = 6.
Line 3: --x makes x = 5 and returns ...

Q47. What is printed? (/ is integer division)
int g(int n) {
if (n == 0) return 0;
if (n % 2 == 1) return g(n - 1) + n;
return g(n / 2) + n / 2;
}
print(g(10))
(a) 14
(b) 17
(c) 15
(d) 12
Answer: **(A)** | *Explanation:* 14. g(1)=g(0)+1=1; g(2)=g(1)+1=2; g(4)=g(2)+2=4; g(5)=g(4)+5=9; g(10)=g(5)+5 = 14 (confirmed by
code)....

Q48. The graph below is stored as an adjacency list, and this iterative traversal marks a node as visited only when it is popped. In what order are the nodes visited?
adj[0] = [1, 2]
adj[1] = [3, 4]
adj[2] = [4, 5]
adj[3] = []
adj[4] = [6]
adj[5] = [6]
adj[6] = []

stack = [0]
while (stack is not empty)
v = pop(stack)
if (v is visited) continue
mark v visited and print v
for each w in adj[v] (in the order listed)
push(stack, w)
end for
end while
(a) 0 2 5 6 4 1 3
(b) 0 1 3 4 6 2 5
(c) 0 1 2 3 4 5 6
(d) 0 2 4 6 5 1 3
Answer: **(A)** | *Explanation:* Visit 0 and push 1, 2. Pop 2, visit, push 4, 5. Pop 5, visit, push 6. Pop 6, visit. Pop 4,

visit (its neighbour
6 is pushed but already vis...

Q49. What is printed?
arr = [3, 4, 10, 11, 14, 7, 0]
count = 0
for i = 0 to 6
for j = i + 1 to 6
if ((arr[i] + arr[j]) % 7 == 0)
count = count + 1
end if
end for
end for
print(count)
(a) 7
(b) 6
(c) 8
(d) 5
Answer: **(A)** | *Explanation:* 7. Taking each value modulo 7 gives 3, 4, 3, 4, 0, 0, 0. A pair sums to a multiple of 7 when it is (3,4) or
(0,0). There are 2×2 = 4 pairs ...

Q50. This is the 0/1 knapsack dynamic programme. What is printed?
wt = [2, 3, 4, 5]
val = [3, 4, 5, 6]
W = 5
dp[0..W] = 0
for i = 0 to 3
for w = W down to wt[i]
dp[w] = max(dp[w], dp[w - wt[i]] + val[i])
end for
end for
print(dp[W])
(a) 7
(b) 6
(c) 8
(d) 9
Answer: **(A)** | *Explanation:* 7. The best choice for capacity 5 is items of weight 2 and 3, worth 3 + 4 = 7. The loop runs w
downwards, so each item is used at most once...

SECTION 5: Numeric Puzzle Solving (10 min)

Q51. Matchstick. A 3 × 3 grid of unit squares is built from 24 matchsticks and contains 14 squares in total (nine 1×1, four 2×2 and one 3×3). What is the minimum number of sticks that must be removed so that exactly 2 complete squares remain?
(a) 4

(b) 3
(c) 5
(d) 6
Answer: **(A)** | *Explanation:* 4. An exhaustive computer search over all 24 sticks shows that no set of 3 sticks leaves exactly 2
squares, while a set of 4 does. Minimum ...

Q52. Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 3, R = 4, B = 5, L = 2, Centre = 9
Wheel 2: T = 6, R = 2, B = 4, L = 5, Centre = 17
Wheel 3: T = 7, R = 3, B = 2, L = 6, Centre = ?
(a) 14
(b) 18
(c) 12
(d) 20
Answer: **(A)** | *Explanation:* 14. Rule: Centre = (T × B) + (R × L) − (T + R + B + L). Wheel 1: 15 + 8 − 14 = 9 ✓. Wheel 2: 24 + 10 − 17 =
17 ✓. Wheel 3: 14 + 18 − 18 = 1...

Q53. Number combination. In the addition SEND + MORE = MONEY, each letter stands for a different digit (0–9), and S and M are not zero. What is the value of D + E + Y?
(a) 14
(b) 12
(c) 16
(d) 13
Answer: **(A)** | *Explanation:* 14. The only solution is S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2 (9567 + 1085 = 10652). D + E + Y = 7 +
5 + 2 = 14....

Q54. Matrix logic. Find the missing number.
3
4
5
5
6
19
7
8
?
(a) 41
(b) 39
(c) 43
(d) 47
Answer: **(A)** | *Explanation:* 41. Rule: third number = (first × second) − (first + second). Row 1: 12 − 7 = 5 ✓. Row 2: 30 − 11 = 19 ✓.

Row 3: 56 − 15 = 41.
Section 6: ...

SECTION 6: English Grammar (10 min)

Q55. A series of failures _____ the board to reconsider the entire strategy.
(a) have led
(b) has led
(c) are leading
(d) were led
Answer: **(B)** | *Explanation:* "A series" is a singular subject, so the verb is "has led"....

Q56. By the time the client called, the team _____ the fix for several hours.
(a) tests
(b) had been testing
(c) has tested
(d) was tested
Answer: **(B)** | *Explanation:* An activity that was in progress for some time before a past event takes the past perfect continuous:
"had been testing"....

Q57. The new policy is incompatible _____ our existing contracts, so we must opt _____ it, or we will be held liable _____ any breach.
(a) with, out of, for
(b) to, out from, of
(c) with, out off, to
(d) to, out of, for
Answer: **(A)** | *Explanation:* incompatible with, opt out of, liable for....

Q58. Error spotting. "The number of vulnerabilities discovered this year (A) / are significantly higher (B) / than that of last year, (C) / according to the report. (D)"
(a) A
(b) B
(c) C
(d) D
Answer: **(B)** | *Explanation:* The subject is "the number" (singular), so the verb should be "is", not "are"....

Q59. She is _____ MBA from _____ university in _____ UK and works as _____ HR manager.
(a) an, a, the, an
(b) a, an, the, a
(c) an, an, -, an
(d) a, a, the, an
Answer: **(A)** | *Explanation:* an MBA (the sound /em/), a university (the sound /ju/), the UK (a specific country name), an HR
manager (the sound /eɪtʃ/).
Section 7: Wri...

SECTION 7: English Writing (10 min)

Q60. Write a short technical essay/paragraph of 150–180 words.
Explain how the "round robin" and "least connections" load-balancing algorithms differ. Then recommend which one is better for a service whose requests vary greatly in how long they take to complete, and justify your recommendation.
Evaluation rubric (10 marks): Technical accuracy of the two algorithms (3) | Soundness and justification of the recommendation (3) | Structure and coherence (2) | Grammar and vocabulary (1) | Word limit adherence (1)
Answer: **(N/A)** | *Explanation:* Evaluation Rubric (10 Marks): Technical accuracy (3) | Comparative analysis (3) | Recommendation justification (2) | Tone & grammar (1) | Wo...